
How Much Can LLMs Hallucinate? An Upper Bound via Coupling and Ambiguity

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Abstract

1 The dominant lineage in hallucination theory bounds *frequency* from below —
2 calibrated LLMs *must* hallucinate at some rate. We bound *size* from above: the goal-
3 conditional displacement of the post-update model state from the goal-marginal,
4 under coupled belief-goal inference. The two axes are orthogonal — how often
5 versus how much an LLM hallucinates. The size axis was open: posterior-stability
6 machinery from Bayesian inverse problems had not been brought to coupled inference.
7

8 The main theorem bounds the displacement by transferred information,

$$\mathbb{E} \|\Delta M_{\text{bias}}\| \leq C \sqrt{I(G; M_{\tau+} \mid e, M_{\tau-})},$$

9 under standard regularity conditions and *independent of any architectural-class*
10 *commitment*. Under a named coupling-attenuation hypothesis, this factors as
11 $C \sqrt{\kappa_{\text{processing}} \cdot I(G; \Omega_{\tau} \mid e_{\tau}, M_{\tau-})}$ — coupling strength times residual ambiguity.
12 Two routes deliver C . A transport-inequality track uses a Talagrand T_2 inequality
13 on the post-update law, verifiable via log-Sobolev concentration or dimension-
14 free sub-Gaussian moments. A Fisher-Rao geometry track yields *two* universal
15 dimension-free constants on d_{FR} : $C = \sqrt{2}$ locally tight under parameterization-
16 invariance + Riemannian structure + KL-coordinate matching + uniform-locality,
17 and $C = \pi/\sqrt{2}$ globally valid under parameterization-invariance alone (with
18 the $\pi/2$ overhead being exactly the worst-case arc-chord ratio at the unit-sphere
19 antipode). The (PI) commitment is load-bearing: no chart-independent universal
20 constant exists under Euclidean coordinates (a two-chart scale-family construction),
21 and the four-condition triple uniquely pins Fisher-Rao + $\sqrt{2}$ as the locally-tight
22 constant. A direct Hellinger statement gives a third universal constant $1/\sqrt{2}$ on the
23 chord — companion to the global Fisher-Rao bound on the same unit L^2 -sphere.

24 Architectures partition by goal-update coupling into a monotonic ladder — Sepa-
25 rated, Partial, Coupled. A structural lemma derives Coupled status for transformer
26 attention from connectivity alone (under non-degenerate attention). For conjugate-
27 Gaussian inverse problems, the bound’s Euclidean translation carries prefactor
28 $L_{\text{post}}\sigma$, bounded by $\tau/2$ uniformly and vanishing in extreme observation-noise
29 limits.

1 Introduction

31 Recent hallucination theory bounds the *frequency* with which calibrated language models must
32 produce false outputs as a consequence of training-set structure or calibration constraints —

33 Kalai and Vempala 2023^[1] on singleton-fact rates, Kalai et al. 2026^[2] on training-and-evaluation-
 34 incentive origins, Karbasi et al. 2025^[3] on detection impossibility, Wu et al. 2024^[4] on VC-
 35 dimension obstructions, Suzuki et al. 2025^[5] on probabilistic negligibility. Hallucination in this
 36 lineage is treated as architecture-agnostic.

37 Bayesian inverse-problems theory in the lineage of Stuart 2010^[6] has spent fifteen years build-
 38 ing Lipschitz-stability bounds on the posterior under perturbations of the prior, the likeli-
 39 hood, and the data — Sprungk 2020^[7], Cvetković and Lie 2025^[8], Dolera and Mainini 2023^[9],
 40 Garbuno-Iñigo et al. 2023^[10], Hosseini et al. 2025^[11]. The transport-inequality apparatus driving
 41 these bounds — Otto-Villani, Bakry-Émery, Gozlan, Bobkov-Götze — is mature and verified across
 42 decades of inverse-problems practice.

43 Neither lineage engages the implicit belief-goal-coupling structure of attention-based architectures —
 44 a structure intrinsic to LLMs by construction (Lemma 3.5), not an incidental property of training.

45 Engaging this structure directly reframes the question: how much can an LLM be pulled away from
 46 what its evidence supports by the goal it is asked to satisfy? This paper bounds the displacement.

47 **This paper composes the two** Apply the chain rule of relative entropy directly to the *post-update*
 48 law, marginalized over the goal: $\mathbb{E}_G[\text{KL}(P_{M_{\tau+}|e, M_{\tau-}, G} \| P_{M_{\tau+}|e, M_{\tau-}})] = I(G; M_{\tau+}|e_{\tau}, M_{\tau-})$.
 49 The right-hand side is the *transferred* goal-information that actually enters the post-update model
 50 state. Combined with a Fisher-Rao second-order expansion on the post-update statistical manifold,
 51 this delivers an upper bound on goal-conditional displacement:

$$\mathbb{E} \|\Delta M_{\text{bias}}\| \leq C \cdot \sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})}. \quad (1)$$

52 Under a parameterization-invariance commitment at full Markov-morphism strength — plus Rieman-
 53 nian structure and KL-coordinate matching — Čencov’s 1982 uniqueness theorem forces Fisher-Rao
 54 with constant $C = \sqrt{2}$. *Universal, dimension-free, no domain-specific parameters*. A scale-family
 55 no-go theorem rules out any coordinate-independent universal constant for Euclidean chart norms —
 56 making the (PI) commitment load-bearing, not coincidental. We call this *Track 2*.

57 A parallel *Track 1* derivation via a Talagrand transport inequality on the same post-update law recovers
 58 the canonical Stuart-school cascade constant $C_{T_2} = 2L_{\text{post}}^2 / \rho_{\text{LSI}}$ as a special case, with $C = \sqrt{C_{T_2}}$
 59 on W_2 . Track 2’s universal constant is what makes the composition novel; Track 1 demonstrates that
 60 the composition contains the existing Bayesian-inverse-problems lineage as a strict sub-case under
 61 standard log-Sobolev concentration. Both flow from the same chain-rule move on the post-update law.

62 **The architectural reading** Under a coupling-attenuation hypothesis (H_{κ}), the bound factors:

$$\mathbb{E} \|\Delta M_{\text{bias}}\| \leq C \cdot \sqrt{\kappa_{\text{processing}} \cdot I(G; \Omega_{\tau} | e_{\tau}, M_{\tau-})}, \quad (2)$$

63 into an *architectural coupling factor* $\kappa_{\text{processing}}$ (how much of the available goal-information the
 64 architecture’s update mechanism transfers) and a *residual ambiguity factor* $I(G; \Omega | e, M)$ (how
 65 much the goal could in principle resolve about the latent world-state beyond what the evidence already
 66 pins down). Architectures partition by their goal-update topology into a monotonic ladder — Class
 67 1 (Separated), Class 2 (Partial), Class 3 (Coupled). For Class 1, $\kappa_{\text{processing}} \leq 1$ holds automatically
 68 by data-processing on the Markov chain $G \rightarrow \Omega \rightarrow M_{\tau+}$. For Class 2 and Class 3, (H_{κ}) is a
 69 named structural commitment with documented sufficient conditions and a worst-case witness (parrot
 70 architecture, $\kappa_{\text{processing}} = \infty$) excluded by scope.

71 Plain decoder-only transformer attention is *Class 3 (Coupled) by construction* in a precise structural
 72 sense: every internal computation has a directed-graph path from any input position whenever at-
 73 tention weights are non-degenerate (Lemma 3.5, by induction on layer depth, robust to RMSNorm /
 74 FlashAttention / causal masking / sliding-window). The structural Coupled-class claim does not de-
 75 pend on the token-distribution-as-belief-state idealization that the bias-bound *application* additionally
 76 invokes via the in-context-learning correspondence^[12–15].

77 **Contributions**

1. **An architectural classification.** Goal/Update Coupling Class — Class 1 (Separated), Class 2 (Partial), Class 3 (Coupled) — partitions architectures monotonically by goal-update topology, grounded in the Pearl-blanket reading^[16] of the Markov-blanket apparatus. Lemma 3.5 derives Class 3 status for transformer attention from connectivity alone.
2. **An umbrella upper bound** (1) on goal-conditional displacement of the post-update model state, derived via chain rule on the post-update law. Two named tracks supply C — a transport-inequality cascade (Track 1) and a Fisher-Rao geometry route (Track 2).
3. **A no-go that forces a coordinate commitment.** No coordinate-independent universal constant exists for Euclidean chart norms (a scale-family construction). Under the (PI) + (R) + (K) triple, Čencov’s uniqueness theorem forces Fisher-Rao + $\sqrt{2}$ universally — no further freedom.
4. **An architectural-corollary form** (2) under (H_κ) , recovering the $\kappa \times \mathcal{A}$ factorization that gives the bound its operational reading.

Section 3 sets up the architectures and the (PI), (R), (K) axioms. Section 4 states the umbrella theorem with both tracks, the no-go, and the architectural corollary. Section 5 sketches the mechanism — both proofs flow from the post-update chain rule. Limitations and the bound’s relation to the frequency-lower-bound lineage are in Section 6.

2 Related Work

Two literatures sit on either side of the gap this paper closes; brief categorization (full positioning across all six adjacent strands in Section F).

LLM hallucination theory has produced architecturally-agnostic frequency lower bounds on calibrated language models — calibration-constrained singleton-fact rates, training-and-evaluation-incentive origins, learnability and computability obstructions, probabilistic-negligibility positive results (cited in §1). These results derive from training-set structure or capacity considerations and do not engage the architecture’s belief-goal-coupling structure. They bound *how often* a calibrated model must hallucinate.

Bayesian inverse-problems posterior stability (the Stuart school) has produced mature Lipschitz-continuity bounds on the posterior under perturbations of the prior, the likelihood, and the data, using transport-inequality machinery (Otto-Villani, Bakry-Émery, Gozlan, Bobkov-Götze) and Fisher-information-flavored constants in optimal-transport frameworks (cited in §1). These results address well-posedness of Bayesian inference; they have not been applied to belief-goal-coupled inference. The machinery bounds *how stable* a Bayesian posterior is to specified perturbations.

The two bound different things, sit in different research traditions, and do not engage with each other. The chain-rule of relative entropy applied directly to the post-update law (Lemma 5.1) is the bridging move: it converts the goal-coupling problem into a transferred-information quantity, which the Stuart-school transport machinery and the Fisher-Rao expansion both bound at the post-update law. *Both* lineages enter — Track 1 inherits the Otto-Villani / Bakry-Émery cascade; Track 2 deploys Čencov uniqueness on the manifold the Stuart school operates on. The composition is the contribution; the architectural reading via Class 1 / Class 2 / Class 3 makes the bound operational.

Adjacent strands — Fisher-Rao geometry for Bayesian sensitivity analysis, Owhadi-Scovel-Sullivan brittleness no-gos, mean-field variational inference stability, the Markov-blanket apparatus from active inference — sit close enough to require explicit cite-and-distinguish but do not resolve the gap. Detailed positioning across all strands is in Section F.

3 Setup

We work with goal-conditioned probabilistic models in a Bayesian-update setting. The state at time τ^- before update consists of an epistemic component M_{τ^-} (the model state, an element of a parameter manifold $\mathcal{M}$) and a goal component G . An event e_τ arrives — a token, an observation, a query response — and the model updates to M_{τ^+} . The latent world-state Ω_τ is what the event is evidence about. The architecture’s update mechanism produces a post-update model state M_{τ^+} whose law depends — possibly stochastically — on the goal G . The goal-conditional post-update law

is $P_{M_{\tau+}|e_{\tau}, M_{\tau-}, G}$; the goal-marginal post-update law $P_{M_{\tau+}|e_{\tau}, M_{\tau-}}$ is the natural baseline against which goal-coupling is measured (the operating-distribution average over goals).

3.1 The bias quantity

The *goal-conditional bias* measures the displacement of the goal-conditional law from the goal-marginal:

$$\|\Delta M_{\text{bias}}(G)\| := \|P_{M_{\tau+}|e_{\tau}, M_{\tau-}, G} - P_{M_{\tau+}|e_{\tau}, M_{\tau-}}\|_{\mathcal{M}}, \quad (3)$$

a random variable in G . Throughout, $\|\cdot\|_{\mathcal{M}}$ denotes either the Wasserstein distance W_2 on the induced model-distribution (Track 1) or the Fisher-Rao geodesic distance d_{FR} on the statistical-manifold sub-case (Track 2). Euclidean-on-parameters is the choice Section 4.1 rules out.

3.2 Goal/Update Coupling Class

Architectures partition by whether G has a causal path into the belief-update computation. The partition is monotonic in coupling strength.

Table 1: Goal/Update Coupling Class — partition by conditional-independence structure.

Class	Goal/update coupling	Examples
Class 1 (Separated)	Holds by construction — estimator has no causal path from G	Kalman filter + LQR; modular RL with separate world model
Class 2 (Partial)	Holds for some pathways, fails for others	Biological cortex; hybrid systems with separate preprocessing
Class 3 (Coupled)	Fails by construction — G is causally upstream of every computation	Transformer LLM (attention processes goals and observations together — Lemma 3.5)

Class 1 (Separated) is the architectural-separation case studied throughout Bayesian inverse-problems theory. Class 3 (Coupled) is the case the existing hallucination-theory lineage doesn't engage geometrically. Class 2 (Partial) is genuinely intermediate — hybrid LLM systems with retrieval, tools, scratchpads, or external memory — and lives at the system level rather than the architecture level. The Pearl-blanket reading^[16] of the Markov-blanket apparatus grounds the partition; we do not adopt the Friston-blanket metaphysical reading.

The bound's architectural-corollary form ((2)) factors transferred goal-information through an *attenuation ratio* — the proportion of in-principle-resolvable goal-information that the architecture's update mechanism actually transfers into the post-update model state:

$$\kappa_{\text{processing}} := \frac{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})}{I(G; \Omega_{\tau} | e_{\tau}, M_{\tau-})}. \quad (4)$$

Conditioning on $M_{\tau-}$ matters: prior correlation between goals and prior model state — present even in Class 1 — would inflate the measure otherwise. For Class 1, the data-processing inequality on the chain $G \rightarrow \Omega_{\tau} \rightarrow M_{\tau+}$ (Markov conditional on $(e_{\tau}, M_{\tau-})$) gives $\kappa_{\text{processing}} \leq 1$ automatically. For Class 2 and Class 3, $\kappa_{\text{processing}} \leq 1$ is a named structural commitment; the parrot architecture $M_{\tau+} := G$ with $G \perp\!\!\!\perp \Omega_{\tau}$ given $(e_{\tau}, M_{\tau-})$ produces $\kappa_{\text{processing}} = \infty$, explicitly outside scope.

3.3 Axioms

The universal-constant route (Section 5.2, Track 2) commits to three axioms on the bound's metric $d_{\mathcal{M}}$:

Hypothesis 3.1 ((PI) Sufficient-statistic invariance). *The bias bound's metric on $\mathcal{M}$ is invariant under sufficient statistics — equivalently, under Markov morphisms / congruent embeddings of statistical manifolds in the sense of Čencov 1982^[17].*

(PI) at full Markov-morphism strength is what Čencov’s uniqueness theorem actually requires. Smooth-coordinate invariance alone admits an infinite family of metrics; the strictly-stronger Markov-morphism class singles out Fisher-Rao up to global scalar.

Hypothesis 3.2 ((R) Riemannian structure). *The bound’s metric $d_{\mathcal{M}}$ derives from a smooth Riemannian metric on $\mathcal{M}$.*

(R) filters out divergence forms (TV, χ^2 , Hellinger-as-divergence) that satisfy (PI) as f -divergences but are not metric distances on $\mathcal{M}$ in the Riemannian sense.

Hypothesis 3.3 ((K) KL second-order matching). $\text{KL}(P_{\theta} \parallel P_{\theta+\delta}) = \frac{1}{2}d_{\mathcal{M}}^2(P_{\theta}, P_{\theta+\delta}) + O(d_{\mathcal{M}}^3)$.

(K) fixes the global scalar Čencov leaves free, pinning the bound’s constant C at $\sqrt{2}$. (R) and (K) are both implicit in any “ $\mathbb{E} d \leq C\sqrt{T}$ ” theorem-shape with second-order matching; we name them explicitly so the uniqueness statement of Theorem 4.5 has its hypotheses fully stated.

The transport-inequality route (Section 5.1, Track 1) and the dimension-restricted slice-wise refinements (Theorem 4.1’s Track 2 locally-tight instantiation) introduce additional regularity hypotheses inline with the theorems that invoke them. Throughout the paper we restrict to the *statistical-manifold sub-case*:

Hypothesis 3.4 ((H1) Statistical-manifold sub-case). *The model state $M_t \in \mathcal{M}$ corresponds to a probability distribution P_{M_t} over latent world-states, and $\mathcal{M}$ is locally a statistical manifold in the sense of Amari and Nagaoka 2000^[18]; $\mathcal{M}$, the goal-space, and the event-space are standard Borel, so regular conditional probabilities exist as Markov kernels^[19] Theorem 6.3.*

(H1) is satisfied by parametric Bayesian posterior families, exponential families, conjugate priors, and the parametric-belief-state subclass of LLMs that admits a natural distributional reading. The standard-Borel regularity is what makes the post-update chain rule (used in both Tracks) hold in the abstract-spaces form^[20] Theorem 3.4; ^[21] Theorem 5.4.

3.4 Coupled-class attention connectivity

The architectural classification’s load-bearing application — that *plain decoder-only transformer attention is structurally Class 3 (Coupled)* — is a property of attention connectivity, not of any token-distribution-as-belief-state idealization.

Lemma 3.5 (Coupled-class attention connectivity). *Let $\mathcal{G}_{\theta}$ be the directed computational graph of an autoregressive transformer with parameters θ , taking input sequence $X_{1:t-1}$ and producing internal activations $\{h_{\ell}^{(i)}\}_{\ell,i}$ across layers ℓ and positions i . If positions $i_G \subseteq \{1, \dots, t-1\}$ contain goal/prompt tokens and positions i_E contain evidence tokens with $i_G \cap i_E = \emptyset$, then for every layer $\ell \geq 1$ and position $j \geq \max(i_G \cup i_E)$, the activation $h_{\ell}^{(j)}$ has a directed-graph path from at least one i_G position whenever attention weights are non-degenerate on the i_G row.*

The proof is a one-induction over layer depth, robust to RMSNorm / GroupNorm / FlashAttention (position-wise / mathematically-equivalent), causal masking (preserves $\text{Attn}_{\ell}^{(j,i_G)} \neq 0$ for $i_G \leq j$), and sliding-window or sparse attention (preserves connectivity by composition across layers when the window pattern admits a multi-hop path). Architectures with explicit goal-to-output attention masking, or token-dropping at intermediate layers, fall outside the lemma’s scope and are appropriately Class 2 (Partial). The full proof is in Section E.5.

Lemma 3.5 establishes a *downstream-output graph-reachability* claim: at every layer and every position causally downstream of the goal and evidence, the activation has a directed-graph path back to at least one goal position. The bias-bound *interpretation* additionally invokes the in-context-learning correspondence^[12–15] between the next-token distribution and an implicit posterior; the structural Coupled-class claim of Lemma 3.5 is robust without it.

4 Main Results

The umbrella theorem bounds the goal-conditional displacement of the post-update model state by transferred goal-information. Two named tracks supply the constant — a transport-inequality cascade (Track 1) and a Fisher-Rao geometry route (Track 2) — both flowing from a common chain-rule move on the post-update law (mechanism in Section 5).

Theorem 4.1 (Umbrella bound on goal-conditional displacement). *Under (H1) and the regularity hypotheses below, for fixed event-prior pair $(e_\tau, M_{\tau-})$ with $\mathbb{E}$ taken over $G \sim P(G \mid e_\tau, M_{\tau-})$:*

$$\mathbb{E} \|\Delta M_{\text{bias}}\| \leq C \cdot \sqrt{I(G; M_{\tau+} \mid e_\tau, M_{\tau-})}, \quad (5)$$

with C supplied by either of two routes:

(Track 1, transport-inequality): *Under additional (H2') — a Talagrand T_2 inequality on the post-update law — $C = \sqrt{C_{T_2}}$ with metric W_2 . The constant C_{T_2} recovers the canonical Stuart-school cascade value $2L_{\text{post}}^2 / \rho_{\text{LSI}}$ as a special case under standard log-Sobolev concentration.*

(Track 2, Fisher-Rao): *Under additional (H4') — uniform-locality on the goal-conditional slice-wise KL — $C = \sqrt{2}$ with metric d_{FR} . The constant is universal, dimension-free, and carries no domain-specific parameters.*

Hypothesis 4.2 ((H2') Talagrand T_2 on the post-update law). *The post-update law $P_{M_{\tau+}|e, M_{\tau-}}$ satisfies $W_2^2(P, P_{M_{\tau+}|e, M_{\tau-}}) \leq C_{T_2} \cdot \text{KL}(P \parallel P_{M_{\tau+}|e, M_{\tau-}})$ for all probability measures P . Sufficient conditions in Section B: log-Sobolev on the post-update density (Otto-Villani $C_{T_2} = 2/\rho_{\text{LSI}}$); dimension-free sub-Gaussian Lipschitz concentration^[22].*

Hypothesis 4.3 ((H4') Uniform local regime). *$\text{ess sup}_g \text{KL}(P_{M_{\tau+}|e, M_{\tau-}, G=g} \parallel P_{M_{\tau+}|e, M_{\tau-}}) \leq \delta_\star$ for some $\delta_\star \in (0, 1)$. Every goal-conditional slice — not merely the average — is uniformly close to the goal-marginal on the statistical manifold. (H4') is strictly stronger than $\mathbb{E}_G[\text{KL}_g] = I \ll 1$; small mean does not imply small ess-sup.*

Remarks on Theorem 4.1.

The two tracks bound different metrics with the same square-root-in-information shape. Track 1 lives on W_2 , Track 2 on d_{FR} . Both deliver bounds of form $\mathbb{E} d \leq C\sqrt{I_M}$ where $I_M = I(G; M_{\tau+} \mid e_\tau, M_{\tau-})$ is transferred goal-information. The two are not “different scalings”; they are bounds on different metrics with the same information dependence.

The bound is unconditional in the architectural classification. Both Tracks bound transferred goal-information directly. (H1)–(H2') and (H1)–(H4')–(PI)–(R)–(K) are regularity hypotheses on the post-update law and the metric. Neither makes a structural commitment about how the goal G enters the architecture's update mechanism. The architectural reading enters at corollary level via (H_κ) below.

Track 2's constant is universal and dimension-free. Track 1's C_{T_2} is family-specific (depends on the post-update concentration). Track 2's $\sqrt{2}$ is the same constant on every statistical manifold under the (PI)+(R)+(K) triple, by Čencov uniqueness (Theorem 4.5 below).

Track 1 contains the Bayesian-inverse-problems lineage. In the conjugate-Gaussian Class 1 (Separated) case with invertible data channel, Track 1's C_{T_2} recovers $2L_{\text{post}}^2 / \rho_{\text{LSI}}$ exactly — the canonical cascade constant from Stuart 2010^[6] and the lineage that follows. The composition contains the existing literature as a strict sub-case.

Track 1 outside (H2'). Heavy-tailed post-updates without log-Sobolev or dimension-free sub-Gaussian concentration fall outside Track 1's T_2 scope. A weaker T_1 form via bounded support gives the distance bound at W_1 rather than W_2 — adequate for the distance reading, weaker on the squared-distance side. Track 2 under (PI)+(R)+(K) does not require (H2') and remains available.

4.1 A no-go that forces (PI)

The universal-constant route (Track 2's $\sqrt{2}$) is not coincidental. A scale-family construction shows that no coordinate-independent universal constant exists for Euclidean chart norms — making the (PI) commitment load-bearing.

Theorem 4.4 (No-go on Euclidean chart norms). *Suppose there were a constant $C_0 < \infty$ — independent of any chart on any statistical manifold — such that for every chart ϕ on $\mathcal{M}$ and every goal-coupled architecture satisfying (H1)+(H2') with non-zero post-update displacement, $\mathbb{E} \|\Delta M_{\text{bias}}\|_{\text{Eucl}, \phi} \leq C_0 \sqrt{I(G; M_{\tau+} \mid e_\tau, M_{\tau-})}$. Then no such C_0 exists.*

The proof is short: a chart rescaling $\phi \mapsto a\phi$ scales the chart-Euclidean Wasserstein distance linearly while leaving the right-hand side (chart-invariant information) unchanged; taking $a \rightarrow \infty$ contradicts the fixed $C_0\sqrt{I}$. Full statement and proof in Section E.3. The Gaussian scale family with chart σ versus $\log \sigma$ is the cleanest illustration.

Remark Theorem 4.4 is *adjacent in shape but opposite in direction* to Owhadi-Scovel-Sullivan’s Bayesian-brittleness results^[23,24]. Their no-go is for *arbitrary stability under finite information* in fixed metrics; ours is for *a universal constant absent a coordinate-invariance commitment*, in a setting where (H2’)’s post-update concentration explicitly excludes their brittleness regime. The two no-gos constrain different things; neither implies the other.

4.2 Čencov uniqueness and sharpness

Track 2’s constant $\sqrt{2}$ is not just universal under (PI)+(R)+(K) — it is the *unique sharp* such constant.

Theorem 4.5 (Uniqueness and sharpness of the Fisher-Rao $+\sqrt{2}$ bound). *Let $\mathcal{M}$ be a statistical-manifold sub-case (H1), and let the bound take the umbrella form $\mathbb{E} d_{\mathcal{M}}(\Delta M_{bias}) \leq C\sqrt{I(G; M_{\tau+} \mid e_{\tau}, M_{\tau-})}$ under the constraints (PI)+(R)+(K). Then:*

(a) $d_{\mathcal{M}}$ is uniquely the Fisher-Rao geodesic distance.

(b) No constant $C < \sqrt{2}$ uniformly bounds $\mathbb{E} d_{FR}/\sqrt{I_M}$ across the (H4’) regime: there exist conjugate-Gaussian Class 1 instances on which $\mathbb{E} d_{FR}^2/I_M \rightarrow 2$ as the goal variation shrinks at fixed I_M . Hence $C = \sqrt{2}$ is the unique sharp upper-bound constant under (PI)+(R)+(K)+(H4’).

Proof in Section E.4. Part (a) is Čencov’s 1982 uniqueness theorem applied to $\mathcal{M}$: the Fisher information metric is the unique (up to global positive scalar) Riemannian metric invariant under Markov morphisms. Part (b) is direct verification on the conjugate-Gaussian saturation.

Remark Alternative chart-independent commitments (TV with Pinsker, Hellinger-as-divergence, χ^2 , Rényi) are all defeated: TV-Pinsker is non-tight (the constant is universal but the inequality is loose except at coincident measures); Hellinger-as-divergence and χ^2 -as-divergence violate (R); switching divergences on the right-hand side violates (K). Under the cascade structure (K) and the metric framing (R), the no-go (Theorem 4.4) tells us *exactly* what to commit to — (PI) — and the universal $\sqrt{2}$ falls out.

4.3 The architectural corollary

The umbrella theorem bounds displacement by *transferred* goal-information. The architectural reading factors this transferred information into a structural-architectural quantity (where can the goal flow?) and an information-theoretic quantity (how much could the goal in principle resolve?).

Hypothesis 4.6 ((H_κ) Bounded architectural attenuation). $\kappa_{processing} \leq \kappa^* < \infty$, with κ^* a named architectural constant (typically 1 for non-amplifying architectures).

Corollary 4.7 (Architectural factorization). *Under (H1) + the Track-specific hypotheses + (H_κ) at level κ^* , the bound factors as*

$$\mathbb{E} \|\Delta M_{bias}\| \leq C\sqrt{\kappa^* \cdot I(G; \Omega_{\tau} \mid e_{\tau}, M_{\tau-})}, \quad (6)$$

with C supplied by Track 1 or Track 2 as in Theorem 4.1.

Remarks on Corollary 4.7.

The factorization is the contribution-headline. Coupling strength $\kappa_{processing} \times$ residual ambiguity $I(G; \Omega \mid e, M)$ — architecture $\times$ information. The unconditional theorem Theorem 4.1 is the structural backbone whose math holds without (H_κ); the corollary is the operational reading.

Class 1 (Separated) gives (H_κ) automatically. The Markov chain $G \rightarrow \Omega_{\tau} \rightarrow M_{\tau+}$ (conditional on $(e_{\tau}, M_{\tau-})$) plus data-processing inequality yields $\kappa_{processing} \leq 1$ by construction. (H_κ) is a derived consequence, not a commitment, for Class 1.

Class 2 (Partial) and Class 3 (Coupled) require (H_κ) as a named structural commitment. Two clean sufficient conditions and a parrot-architecture worst-case witness ($\kappa_{\text{processing}} = \infty$ when $M_{\tau+} := G$ on goals orthogonal to the latent world-state) are documented in Section B.3. The unconditional theorem still bounds parrot-architecture displacement directly by transferred information; the $\kappa \times \mathcal{A}$ factorization just doesn't apply.

The realized $\kappa_{\text{processing}}$ is empirically accessible. A behavioral two-goal probe — run the architecture on a fixed event set under two goal states, measure divergence of the epistemic content of the response — gives a lower bound on $\kappa_{\text{processing}}$ at the operating point. For architectures with explicit attention or routing, activation-level mediation analyses give a tighter handle. Both are out of scope for the present theory paper.

5 Mechanism

Both tracks bridge to the bound through the same move: a chain rule of relative entropy applied directly to the post-update law, marginalized over the goal. This is the structural composition of the two literatures — the Bayesian-inverse-problems posterior-stability machinery and the architectural classification meet at this identity.

Lemma 5.1 (Chain rule on the post-update law). *Under (H1),*
 $\mathbb{E}_G[\text{KL}(P_{M_{\tau+}|e, M_{\tau-}}, G \parallel P_{M_{\tau+}|e, M_{\tau-}})] = I(G; M_{\tau+} \mid e_\tau, M_{\tau-}).$

The identity is exact. The right-hand side is *transferred* goal-information — the actually-realized coupling between G and the post-update model state. The left-hand side averages KL between goal-conditional and goal-marginal post-update laws. (H1)'s standard-Borel regularity is what makes this identity hold in the abstract-spaces form (Polyanskiy and Wu²⁰, Theorem 3.4; Gray²¹, Theorem 5.4). Combined with either a transport inequality on the post-update law or a Fisher-Rao second-order expansion of KL, the bound follows by taking expectation over G and applying Jensen.

5.1 Track 1 — transport-inequality cascade

Two steps.

Step 1 Apply Lemma 5.1: $\mathbb{E}_G[\text{KL}_g] = I_M$ where KL_g is the goal-conditional-vs-marginal KL at $G = g$ and $I_M = I(G; M_{\tau+} \mid e_\tau, M_{\tau-})$.

Step 2 Apply (H2') slice-wise: $W_2^2(P_{M_{\tau+}|G=g}, P_{M_{\tau+}}) \leq C_{T_2} \cdot \text{KL}_g$. Take expectation over G and substitute Step 1:

$$\mathbb{E}[W_2^2(P_{M_{\tau+}|G}, P_{M_{\tau+}})] \leq C_{T_2} \cdot I_M.$$

Jensen on $\sqrt{\cdot}$ delivers the distance form $\mathbb{E} W_2 \leq \sqrt{C_{T_2} I_M}$.

The cascade is composed from textbook information-theoretic and transport-inequality machinery — Otto-Villani for the LSI-implies- T_2 direction, Bakry-Émery for the K -strong-log-concavity-implies-LSI direction, Gozlan's characterization of T_2 via dimension-free sub-Gaussian Lipschitz concentration, Bobkov-Götze for the strictly weaker single-Lipschitz-function T_1 version. The contribution is the application of these machinery elements to the goal-conditional bias quantity on the post-update law: there is no separate posterior-pushforward step, because the chain rule lives on the post-update law directly. Full proof and (H2') sufficient-condition verifications in Section B.

5.2 Track 2 — Fisher-Rao expansion

One step.

Lemma 5.2 (KL-to-Fisher-Rao second-order expansion). *On a statistical manifold with Fisher information metric $\mathbf{I}$, for nearby P, Q : $\text{KL}(P \parallel Q) = \frac{1}{2} d_{FR}^2(P, Q) + O(d_{FR}^3)$.*

This is the infinitesimal form of the Bregman divergence on the exponential-family Fenchel geometry — exact at second order, tight in the small-information regime (Cover and Thomas²⁵ §12.5; Amari

and Nagaoka¹⁸ §3.7 Theorem 3.1). Apply Lemma 5.2 slice-wise at each $G = g$ under (H4') (every goal-conditional slice uniformly close to the goal-marginal on the manifold, so the expansion is sharp slice-wise with controlled remainder). Take expectation over G and substitute Lemma 5.1:

$$\mathbb{E}[d_{FR}^2(P_{M_{\tau+}|G}, P_{M_{\tau+}})] \leq 2 \cdot I_M \cdot (1 + o(1)).$$

Jensen on $\sqrt{\cdot}$: $\mathbb{E} d_{FR} \leq \sqrt{2} \sqrt{I_M} (1 + o(1))$.

The $(1 + o(1))$ is the third-order remainder, controlled uniformly under (H4') by the Amari-Chentsov tensor and the minimum Fisher eigenvalue along the goal-induced geodesic. The constant $\sqrt{2}$ is what (PI)+(R)+(K) buys: Čencov uniqueness (Theorem 4.5) forces Fisher-Rao with that exact constant.

Outside (H4') When rare goals carry disproportionately large transferred information, or when the operating distribution sits at moderate-to-large I_M , the slice-wise expansion is no longer uniformly sharp. Track 2 has globally-valid companion bounds — Fisher-Rao at $\pi/\sqrt{2}$ via the unit L^2 -sphere chord-arc bound, and a direct Hellinger statement at $1/\sqrt{2}$ — under (H1)+(PI) only, no (R), (K), or (H4'). The full taxonomy and proofs are in Section D. The $\pi/2 \approx 1.57$ overhead is the worst-case arc-chord ratio at the unit L^2 -sphere's antipode — a geometric maximum, not a slack.

5.3 No-go — chart rescaling

The proof of Theorem 4.4 is short. Under any chart ϕ on $\mathcal{M}$ and rescaling $\phi_a := a\phi$ (a smooth coordinate change the underlying geometry treats as identity):

- *Chart-Euclidean W_2 scales linearly*: $W_2^{\phi_a}(\mu, \nu) = a \cdot W_2^{\phi}(\mu, \nu)$ — the pushforward under $x \mapsto ax$ rescales transport cost by a .
- *KL, conditional MI, Fisher-Rao geodesic, Hellinger — all chart-invariant*: relative entropies depend only on densities and dominating measures (reparameterization-invariant); Fisher metric tensor transforms covariantly so integrated geodesic distance is invariant; Hellinger as f -divergence is reparameterization-invariant.

For any architecture realizing finite positive $W_2^{\phi} > 0$ and finite positive transferred information I , the candidate universal-constant inequality would have to satisfy $a \cdot \|\Delta M_{\text{bias}}\|_{\text{Eucl}, \phi} \leq C_0 \sqrt{I}$ for all $a > 0$ at fixed $C_0 \sqrt{I}$. Taking $a \rightarrow \infty$ contradicts the fixed right-hand side. Both required quantities ($W_2^{\phi} > 0$ and $I > 0$) are automatic on any non-trivial goal-coupled architecture — for instance, the conjugate-Gaussian Class 1 setup with any $\sigma, \tau > 0$ and a non-degenerate continuous goal distribution. The Gaussian scale family with two charts (Chart A: σ ; Chart B: $\log \sigma$) is the canonical illustration: the same intrinsic geometry, Chart-A Euclidean displacement grows linearly with the operating scale while Chart-B stays bounded.

The implication is constructive. To get a universal constant the bound must be stated in a coordinate-invariant norm — one that respects the underlying geometry of the statistical manifold rather than an arbitrary chart on it. This is exactly the role Fisher-Rao plays under (PI)+(R)+(K). Alternative chart-independent commitments (TV with Pinsker, Hellinger-as-divergence, χ^2 , Rényi) yield strictly weaker bounds — the no-go does not just say “commit to something intrinsic”; it says *exactly* what to commit to.

6 Conclusion

The opening question — *how much can an LLM be pulled away from what its evidence supports by the goal it is asked to satisfy?* — is the question the existing hallucination-theory lineage doesn't engage. The lineage bounds *frequency*; the question is about *size*. The two axes are orthogonal. A coupled-architecture model in (PI)-compatible operating regime can simultaneously have an inevitable-frequency hallucination floor (training-data structure, calibration constraints, learnability bottlenecks, computability obstructions) *and* a bounded-magnitude goal-coupling displacement on each event (geometric structure). The two together give a more complete picture than either alone.

What the geometric route delivers — and what the architecture-engaged commitment buys. The chain-rule move on the post-update law composes two literatures that hadn’t met. Track 2 under (PI)+(R)+(K) gives a universal, dimension-free $\sqrt{2}$ constant by Čencov uniqueness; the no-go on Euclidean chart norms makes that commitment load-bearing rather than coincidental. The architectural classification reads the bound operationally: $\kappa_{\text{processing}} \times \mathcal{A}$ — coupling strength times residual ambiguity — with Class 1 (Separated) automatic by data-processing, Class 2 (Partial) and Class 3 (Coupled) named structural commitments. Plain decoder-only transformer attention is structurally Class 3 by Lemma 3.5, not as an incidental property of training.

Limitations The bounds are conditional theorems. Regularity hypotheses (H2’) and (H4’) restrict the regime each Track applies in; (PI) is an axiomatic commitment, not a derivation; the architectural factorization requires (H_κ) at corollary level. The paper is theory-only — we do not instrument a deployed LLM’s belief-update dynamics. The mitigation we offer is structural: Track 1’s cascade has decades of empirical use in Bayesian inverse problems; Track 2 under (PI)+(R)+(K) is mathematically airtight via Čencov; the no-go is counterexample-grade. The bound captures *one channel* of hallucination — goal-coupling-induced displacement — which is real and structurally derived. Other channels (training-data structure, calibration, capacity) live in the orthogonal frequency lineage and are addressed there.

Future directions Empirical estimation of $\kappa_{\text{processing}}$ for deployed LLMs via two-goal probing or activation-level mediation analysis; extension of the locally-tight Fisher-Rao bound to the moderate-to-large I regime via compact-manifold or staged-composition arguments; instantiation of the bound for specific architecture families (mixture-of-experts, retrieval-augmented, state-space LLMs that fall outside Lemma 3.5’s attention-based scope); and connection to the frequency-lower-bound lineage of Kalai-Vempala-style results to give a layered hallucination theory in which structural-bias bounds at the per-event level compose with frequency bounds at the corpus level.

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A Failed routes

We document two derivation routes that failed at structural levels, so future work does not re-attempt them without new evidence.

A.1 Route F1: Cramér-Rao inversion

Attempt Treat G as a parameter to be estimated from Ω given (e, M) . The Cramér-Rao lower bound gives $\text{Var}(\hat{G}) \geq \mathbf{I}_G^{-1}$ where $\mathbf{I}_G$ is the Fisher information about G given (Ω, e, M) . Invert this to a posterior-sensitivity factor: an upper bound $\|\Delta M_{\text{bias}}\| \leq L_M \cdot \sqrt{\mathbf{I}_G^{-1}}$ via a Lipschitz-in- G constant on the coupled-update mechanism.

Why it fails Cramér-Rao bounds estimator variance from *below*. To produce an *upper bound* on $\|\Delta M_{\text{bias}}\|$ via Cramér-Rao inversion would require fixing an estimator class within which $\mathbf{I}_G^{-1}$ is also an upper bound on variance — but no such estimator class exists for the coupled architecture’s update mechanism. The mechanism is a deterministic function of (G, e, M) , not an estimator of G , and there is no uniform-class assumption available. The direction of the Cramér-Rao bound and the direction of the bias bound are opposite; inverting one to the other requires an additional assumption that is not available.

Lesson Information-theoretic *lower-bound* machinery for estimator performance does not invert to upper bounds on side-information-induced displacement.

A.2 Route F2: Rate-distortion inversion

Attempt Apply Shannon’s rate-distortion inequality $R(D) \geq h(X) - \frac{1}{2} \log(2\pi e D)$ for a Gaussian source. Invert to $D \leq \sigma^2 \cdot 2^{-2(R-h(X))}$ with “bits in” interpreted as $I(G; \Omega|e, M)$, “distortion” interpreted as $\|\Delta M_{\text{bias}}\|^2$.

Why it fails Rate-distortion theory describes the minimum bits-per-symbol required to *represent* a source within distortion D ; it does not describe the maximum *displacement induced* by injecting side-information into an update mechanism. The two problems have different structures: source-coding is about optimal representations; side-information injection is about perturbation propagation. Identifying “bits in” with I and “distortion” with displacement does not produce a sound inequality — the rate-distortion curve is about a different operational quantity.

565 **Lesson** Source-coding theorems are about optimal representations of a source, not spatial dis-
 566 placement induced by side-information injection. Upper bounds on displacement require transport-
 567 inequality machinery (Pinsker, Otto-Villani, Bakry-Émery, Lipschitz-posterior).

568 A.3 The pattern

569 Both failed routes attempt to invert *information-theoretic lower-bound* machinery (Cramér-Rao
 570 for estimator error; rate-distortion for source coding) into *upper bounds on displacement*. The
 571 structural mismatch is uniform: lower-bound machinery for optimization quantities does not invert to
 572 upper bounds on perturbation quantities without additional structure that is not available in coupled-
 573 architecture settings. Transport-inequality machinery is on the other side of this divide and is the
 574 correct family for upper-bound-on-displacement results.

575 B Hypothesis verification details

576 This appendix collects the regularity-condition verifications referenced in Section 5.1 and Section 5.2.

577 B.1 (H2') Talagrand T_2 inequality on the post-update law — sufficient conditions

578 (H2') requires $W_2^2(P, P_{M_{\tau+}|e, M_{\tau-}}) \leq C_{T_2} \text{KL}(P \| P_{M_{\tau+}|e, M_{\tau-}})$ for all probability measures P .
 579 Three clean sufficient conditions:

580 **Route (a): LSI on the post-update law** If $P_{M_{\tau+}|e, M_{\tau-}}$ satisfies a logarithmic Sobolev inequality
 581 with constant $\rho_{\text{post}} > 0$,

$$\text{Ent}_{P_{M_{\tau+}|e, M_{\tau-}}}(f^2) \leq \frac{2}{\rho_{\text{post}}} \mathbb{E}_{P_{M_{\tau+}|e, M_{\tau-}}} [\|\nabla f\|^2] \quad \text{for all sufficiently smooth } f,$$

582 then by^[26] Theorem 1, (H2') holds with $C_{T_2} = 2/\rho_{\text{post}}$. A sufficient condition for LSI, by Bakry-
 583 Émery's curvature-dimension criterion^[27], is K -strong log-concavity of the post-update density:
 584 $-\nabla^2 \log p_{M_{\tau+}|e, M_{\tau-}} \succeq K \cdot I$ for some $K > 0$, giving $\rho_{\text{post}} \geq K$.

585 **Hypothesis (S) — own proof of strong log-concavity for Bayesian post-updates** When the post-
 586 update law is itself a Bayesian posterior with negative log-likelihood $\Phi(\theta; \omega)$ and prior density $\pi(\theta)$,
 587 the posterior negative log-density is $U(\theta; \omega) = \Phi(\theta; \omega) - \log \pi(\theta)$. Strong log-concavity of the
 588 posterior is a *direct* statement about the Hessian-in- θ of U : $\nabla_{\theta}^2 U(\theta; \omega) \succeq K_{\text{eff}} \cdot I$ uniformly in $\omega \in \Omega$,
 589 for some $K_{\text{eff}} > 0$. Two clean sufficient conditions:

- 590 • (S-i) *Convex likelihood + strongly log-concave prior*. If $\nabla_{\theta}^2 \Phi(\theta; \omega) \succeq 0$ uniformly in (θ, ω)
 591 (likelihood log-concave in θ) and π is K -strongly log-concave (so $-\nabla_{\theta}^2 \log \pi \succeq K \cdot I$), then
 592 $\nabla_{\theta}^2 U \succeq K \cdot I$, giving $K_{\text{eff}} \geq K$.
- 593 • (S-ii) *Bounded negative likelihood curvature dominated by prior curvature*. If $\nabla_{\theta}^2 \Phi(\theta; \omega) \succeq$
 594 $-K_0 \cdot I$ uniformly with $K_0 < K$ (likelihood may be locally non-convex but its negative-
 595 curvature is bounded), and π is K -strongly log-concave, then $\nabla_{\theta}^2 U \succeq (K - K_0) \cdot I$, giving
 596 $K_{\text{eff}} \geq K - K_0 > 0$.

597 Either condition gives Bakry-Émery $\rho_{\text{post}} \geq K_{\text{eff}}$ and Otto-Villani $C_{T_2} \leq 2/K_{\text{eff}}$. Sprungk 2020^[7]
 598 derives sharper local-Lipschitz constants for the posterior across TV / Hellinger / W_2 / KL;
 599 Cvetković and Lie 2025^[8] establish matched lower bounds; Dolera and Mainini 2023^[9] give global
 600 W_2 -Lipschitz constants in terms of Fisher-information functionals — which can sharpen C_{T_2} beyond
 601 the bare $2/K_{\text{eff}}$. Mixed-Hessian smoothness $\|\partial_{\theta\omega}^2 \Phi\| \leq H$ is *not* a substitute: it controls how the
 602 posterior mode shifts with ω , but the LSI / strong-convexity claim is about the curvature in θ and
 603 must be made directly.

604 **Route (b) (corrected — bounded support gives T_1 , not T_2)**. Bare bounded support of diameter D on $\mathcal{M}$
 605 is **not** a T_2 -sufficient condition. *Counterexample*. On $\{0, 1\} \subset \mathbb{R}$ (diameter $D = 1$), let $\nu = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$
 606 and $\mu_{\epsilon} = (\frac{1}{2} - \epsilon)\delta_0 + (\frac{1}{2} + \epsilon)\delta_1$ for $\epsilon \in (0, 1/2)$. Then $W_2^2(\mu_{\epsilon}, \nu) = \epsilon$ (the optimal coupling moves
 607 mass ϵ from 0 to 1), while $\text{KL}(\mu_{\epsilon} \| \nu) = 2\epsilon^2 + O(\epsilon^4)$, so $W_2^2/\text{KL} = 1/(2\epsilon) + O(1) \rightarrow \infty$ as

608 $\epsilon \rightarrow 0^+$ — no constant C uniformizes the ratio. The structural reason: Gozlan 2009^[22] established
 609 that T_2 is *equivalent* to dimension-free Gaussian concentration of $\mu^{\otimes n}$, strictly stronger than diameter
 610 (Talagrand-type product concentration requires LSI / curvature input, not just bounded support).
 611 What does follow from diameter is the *weaker* T_1 inequality: by $W_1 \leq D \cdot \text{TV}$ and Pinsker $\text{TV} \leq$
 612 $\sqrt{\text{KL}/2}$,

$$W_1(P, P_{M_{\tau+}|e, M_{\tau-}}) \leq \frac{D}{\sqrt{2}} \sqrt{\text{KL}(P \| P_{M_{\tau+}|e, M_{\tau-}})}, \quad C_{T_1} = D^2/4,$$

613 sharp on the two-point Bernoulli at $p = 1/2$. (Equivalently, this is Bobkov and Götze²⁸ with the
 614 Hoeffding sub-Gaussian variance proxy $\sigma^2 = D^2/4$ for distributions on a diameter- D set, applied to
 615 the T_1 -equivalent Lipschitz-sub-Gaussian characterization.) Substituting W_1 for W_2 in Step 2 of the
 616 Section 5.1 cascade preserves the *distance-form* bound $\mathbb{E}\|\Delta M_{\text{bias}}\| \leq (D/\sqrt{2})\sqrt{I(G; M_{\tau+}|e, M_{\tau-})}$
 617 at the T_1 level — at the cost of giving up Theorem 4.1’s tighter squared-distance / linear-in- I refinement
 618 (5). Token-distribution belief states for LLMs on the simplex are more naturally treated under Track 2
 619 (Fisher-Rao, Section 5.2), where the simplex’s constant positive sectional curvature (on the round-
 620 sphere image $x \mapsto (\sqrt{x_1}, \dots, \sqrt{x_n})$) provides Bakry-Émery $\text{CD}(K, n-1)$ with $K > 0$, an LSI w.r.t.
 621 Fisher-Rao volume, and the universal $\sqrt{2}\sqrt{I}$ form holds without Track 1’s diameter-only weakening.

622 **Route (c): dimension-free sub-Gaussian Lipschitz concentration** Per Gozlan 2009^[22], μ satisfies
 623 $T_2(C)$ iff $\mu^{\otimes n}$ enjoys dimension-free Gaussian concentration of Lipschitz functionals: there exist
 624 constants $C', \lambda > 0$ such that $\mathbb{E}_{\mu^{\otimes n}}[\exp(\lambda(f - \mathbb{E}f)^2)] \leq 2$ uniformly in n for every 1-Lipschitz
 625 $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Bobkov and Götze 1999^[28] characterizes the strictly weaker *single-Lipschitz-function*
 626 version (giving T_1); the Gozlan strengthening to T_2 is what makes single-distribution sub-Gaussian
 627 moments insufficient on their own. When dimension-free product concentration holds, C_{T_2} admits an
 628 explicit form in the variance proxy.

629 **When both routes fail** Heavy-tailed post-updates without LSI and without dimension-free sub-
 630 Gaussian concentration are outside Track 1’s T_2 scope. The T_1 fallback under bounded support (Route
 631 (b) as corrected above) recovers the distance form at weaker constants — adequate for $\mathbb{E}\|\Delta M_{\text{bias}}\| \leq$
 632 $C\sqrt{I}$. Track 2 under (PI) + (R) + (K) does not require (H2') and remains available; Theorem D.2’s
 633 Hellinger backstop is a global universal-constant fallback under (PI) only.

634 B.2 KL-to-Fisher second-order expansion

635 For nearby distributions $P_\theta, P_{\theta+\delta}$ on a parametric family with Fisher information $\mathbf{I}(\theta)$:

$$\text{KL}(P_{\theta+\delta} \| P_\theta) = \frac{1}{2} \delta^\top \mathbf{I}(\theta) \delta + O(\|\delta\|^3) = \frac{1}{2} d_{FR}^2(P_\theta, P_{\theta+\delta}) + O(d_{FR}^3).$$

636 Standard derivation: Cover and Thomas²⁵ §12.5; Amari and Nagaoka¹⁸ §3.7 Theorem 3.1. The
 637 $O(d_{FR}^3)$ remainder is sharp in the sense that the second-order term is exact and the third-order term
 638 is generically non-zero (off-exponential-family).

639 B.3 $(\mathbf{H}_\kappa)$ sufficient conditions

640 The architectural-corollary commitment of Section B.3 is that $\kappa_{\text{processing}} =$
 641 $I(G; M_{\tau+}|e, M_{\tau-})/I(G; \Omega_\tau|e, M_{\tau-})$ is bounded by some named architectural constant $\kappa^* < \infty$
 642 (typically 1). For Class 1 (Separated) architectures, the data-processing inequality applied to the
 643 Markov chain $G \rightarrow \Omega_\tau \rightarrow M_{\tau+}$ (conditional on $(e_\tau, M_{\tau-})$) by architectural separation) gives
 644 $\kappa_{\text{processing}} \leq 1$ automatically. For Class 2 (Partial) and Class 3 (Coupled), two structural conditions
 645 each give $\kappa_{\text{processing}} \leq 1$ (or $\leq 1 + \alpha$ in the second condition):

- 646 1. *Goal-blind effective kernel.* Suppose the post-update map factors as $M_{\tau+} =$
 647 $K(M_{\tau-}, e_\tau, \tilde{\Omega}_\tau)$ with K deterministic, where $\tilde{\Omega}_\tau$ is a goal-conditional reweighting of
 648 Ω_τ that is *non-amplifying*: $I(G; \tilde{\Omega}_\tau|e_\tau, M_{\tau-}) \leq I(G; \Omega_\tau|e_\tau, M_{\tau-})$. Then conditional
 649 on $(e_\tau, M_{\tau-})$, $G \rightarrow \tilde{\Omega}_\tau \rightarrow M_{\tau+}$ is a Markov chain (since K depends on G only through
 650 $\tilde{\Omega}_\tau$), and DPI applied along this chain gives

$$I(G; M_{\tau+}|e_{\tau}, M_{\tau-}) \leq I(G; \tilde{\Omega}_{\tau}|e_{\tau}, M_{\tau-}) \leq I(G; \Omega_{\tau}|e_{\tau}, M_{\tau-}),$$

i.e., $\kappa_{\text{processing}} \leq 1$.

1. *Bounded direct architectural channel.* Suppose the architecture’s *direct* goal-coupling pathway (independent of Ω_{τ}) has bounded conditional MI relative to the resolvable channel: $I(G; M_{\tau+}|e_{\tau}, M_{\tau-}, \Omega_{\tau}) \leq \alpha I(G; \Omega_{\tau}|e_{\tau}, M_{\tau-})$ for some $\alpha \geq 0$. The chain rule of mutual information conditioned on $(e_{\tau}, M_{\tau-})$ gives $I(G; M_{\tau+}, \Omega_{\tau}|e_{\tau}, M_{\tau-}) = I(G; \Omega_{\tau}|e_{\tau}, M_{\tau-}) + I(G; M_{\tau+}|e_{\tau}, M_{\tau-}, \Omega_{\tau})$, and using $I(G; M_{\tau+}|e_{\tau}, M_{\tau-}) \leq I(G; M_{\tau+}, \Omega_{\tau}|e_{\tau}, M_{\tau-})$ (additional side information cannot decrease MI), we get $\kappa_{\text{processing}} \leq 1 + \alpha$. The case $\alpha = 0$ — architectural channel adds nothing beyond what Ω_{τ} provides given $(e_{\tau}, M_{\tau-})$ — gives $\kappa_{\text{processing}} \leq 1$.

Worst-case witness — parrot architecture Consider a degenerate Coupled-class architecture whose update rule is $M_{\tau+} := G$ (the post-update model state is identically the goal). Two computations:

Numerator. Since $M_{\tau+} = G$ deterministically conditional on $(e_{\tau}, M_{\tau-})$, $I(G; M_{\tau+}|e_{\tau}, M_{\tau-}) = H(G|e_{\tau}, M_{\tau-})$ exactly.

Denominator. Suppose $G \perp\!\!\!\perp \Omega_{\tau}$ given $(e_{\tau}, M_{\tau-})$ — the goal carries no information about the latent world-state beyond what the evidence and prior have already pinned down. (Operationally this is the “make me feel good” goal of Section 3.2: the user’s affective goal is independent of the empirical question the evidence resolves.) Then $I(G; \Omega_{\tau}|e_{\tau}, M_{\tau-}) = 0$.

Hence $\kappa_{\text{processing}} = H(G|e_{\tau}, M_{\tau-})/0 = +\infty$. (H_{κ}) explicitly fails. This shows the architectural commitment is *real* for non-Separated architectures, not a free move. The unconditional theorems (Theorem 4.1, Theorem 4.1, Theorem D.2) still bound the parrot architecture’s displacement by transferred information directly — the prefactor is $\sqrt{H(G|e_{\tau}, M_{\tau-})}$ rather than $\sqrt{\kappa I(G; \Omega)}$ — but the $\kappa \times \mathcal{A}$ factorization (and hence the architectural-corollary form) does not apply.

C Conjugate-Gaussian numerical comparison

We compute the three tracks’ constants explicitly for the conjugate-Gaussian observation-prior pair and exhibit the regimes in which they are tight. The setup is the one introduced in Section C: latent $\theta \sim \mathcal{N}(0, \tau^2)$, observation channel $\Omega|\theta \sim \mathcal{N}(\theta, \sigma^2)$, goal-conditional likelihood-mean shift $\Omega|G \sim \mathcal{N}(\beta(G), \sigma^2)$. The architecture’s update is goal-blind (Class 1 Separated, conjugate Bayesian update). Identifying $M_{\tau+}$ with the posterior-mean parameter $\mu_{+} := L_{\text{post}}\Omega$ (variance $\sigma_{\text{post}}^2 = L_{\text{post}}\sigma^2$ fixed), the post-update parameter law is $\mu_{+}|G \sim \mathcal{N}(L_{\text{post}}\beta(G), L_{\text{post}}^2\sigma^2)$.

Two distinct variances. The *inside-posterior covariance* $\sigma_{\text{post}}^2 = L_{\text{post}}\sigma^2$ is the spread of the posterior measure on θ at a fixed observation. The *parameter-law variance* $L_{\text{post}}^2\sigma^2 = L_{\text{post}}\sigma_{\text{post}}^2$ is the spread of the random model state μ_{+} under data noise. They differ by exactly L_{post} . Track 1’s C_{T_2} is determined by the parameter-law variance, not the inside-posterior covariance; conflating them is the algebraic error this appendix corrects.

Transferred information and $\kappa_{\text{processing}}$. The goal-conditional law of Ω has small- $\text{Var}_G(\beta)$ MI $I(G; \Omega|e, M_{\tau-}) \approx \text{Var}_G(\beta)/(2\sigma^2)$. The post-update parameter law inherits the same MI: the deterministic linear map $\Omega \mapsto L_{\text{post}}\Omega$ is invertible, so the data-processing inequality on the chain $G \rightarrow \Omega \rightarrow M_{\tau+}$ saturates and $I(G; M_{\tau+}|e, M_{\tau-}) = I(G; \Omega|e, M_{\tau-})$ exactly. The realized attenuation ratio is $\kappa_{\text{processing}}^{\text{realized}} = 1$. *This Class 1 conjugate-Gaussian setup does not exhibit architectural attenuation* — it sits at the worst-case Class 1 corner where DPI is tight; non-trivial $\kappa < 1$ requires a non-invertible data channel (e.g., quantization, dimension reduction) or a Class 2/3 architecture.

Track 1 (Theorem 4.1). Route (a) — LSI on the post-update parameter law $\mathcal{N}(0, L_{\text{post}}^2\sigma^2)$ at small $\text{Var}_G(\beta)$ — gives $\rho_{\text{post}} = 1/(L_{\text{post}}^2\sigma^2)$ and

$$C_{T_2} = 2/\rho_{\text{post}} = 2L_{\text{post}}^2\sigma^2 = \frac{2\tau^4\sigma^2}{(\sigma^2 + \tau^2)^2}.$$

695 Theorem 4.1: $\mathbb{E}[W_2^2] \leq C_{T_2} \cdot I(G; M_{\tau+}|e, M_{\tau-}) = 2L_{\text{post}}^2\sigma^2 \cdot I(G; \Omega|e, M_{\tau-})$ (using $\kappa = 1$).
696 Equivalently, with $\rho_{\text{LSI}} := 1/\sigma^2$,

$$\mathbb{E}[W_2^2] \leq \frac{2L_{\text{post}}^2}{\rho_{\text{LSI}}} \cdot I(G; \Omega|e, M_{\tau-})$$

697 — the canonical Stuart-school cascade constant, recovered as C_{T_2} alone (DPI saturation $\kappa = 1$ is the
698 witness that this Class 1 example sits at the Stuart corner). The cascade is sharp: the slice-wise W_2^2/KL
699 ratio equals C_{T_2} exactly for these Gaussian parameter laws. Limit checks: prior-dominant ($\tau \ll \sigma$):
700 $C_{T_2} \approx 2\tau^4/\sigma^2 \rightarrow 0$; likelihood-dominant ($\sigma \ll \tau$): $C_{T_2} \approx 2\sigma^2 \rightarrow 0$; balanced ($\tau = \sigma = 1$):
701 $C_{T_2} = 1/2$ and $\mathbb{E}[W_2^2] \leq I_M/2$.

702 **Track 2 (Theorem 4.1).** Fisher metric on the parameter-law manifold $\{\mathcal{N}(\mu, L_{\text{post}}^2\sigma^2) : \mu \in \mathbb{R}\}$
703 (laws over the random parameter μ_+ , not posterior measures on θ): $\mathbf{I}(\mu) = 1/(L_{\text{post}}^2\sigma^2)$. Fisher-
704 Rao distance $d_{FR}(\mathcal{N}(\mu_1, v), \mathcal{N}(\mu_2, v)) = |\mu_1 - \mu_2|/\sqrt{v} = |\mu_1 - \mu_2|/(L_{\text{post}}\sigma)$. Slice-wise to
705 the marginal $\mathcal{N}(0, L_{\text{post}}^2\sigma^2)$: $d_{FR}(P_{M_{\tau+}|G=g}, P_{M_{\tau+}}) \approx |\beta(g)|/\sigma$ at small $\text{Var}_G(\beta)$. Theorem 4.1
706 (unconditional): $\mathbb{E}[d_{FR}^2] \leq 2 \cdot I(G; M_{\tau+}|e, M_{\tau-}) = \text{Var}_G(\beta)/\sigma^2$ — sharp on this family.

707 **Track 2 Euclidean translation (Theorem D.4).** $|\Delta\mu_+|_{\text{Eucl}} = L_{\text{post}}\sigma \cdot d_{FR}$, so $\mathbb{E}|\Delta\mu_+|_{\text{Eucl}} \leq$
708 $L_{\text{post}}\sigma\sqrt{2I_M}$. The prefactor $L_{\text{post}}\sigma = \tau^2\sigma/(\sigma^2 + \tau^2)$ is bounded above by $\tau/2$ uniformly (AM-GM
709 at $\sigma = \tau$) and *vanishes in both extreme limits*: $L_{\text{post}}\sigma \rightarrow 0$ as $\sigma \rightarrow \infty$ (very noisy observations: the
710 architecture learns essentially nothing from Ω , posterior-mean displacement vanishes); $L_{\text{post}}\sigma \rightarrow 0$ as
711 $\sigma \rightarrow 0$ (sharp observations: $L_{\text{post}} \rightarrow 1$ but $\sigma \rightarrow 0$). Maximum at balanced $\sigma = \tau$. The naive “ $\sigma\sqrt{2I}$ ”
712 reading is wrong (correct prefactor is $L_{\text{post}}\sigma$, not bare σ); equally importantly, the “ $\sigma_{\text{post}}\sqrt{2I}$ ” reading
713 is *also* wrong (it conflates inside-posterior covariance $\sigma_{\text{post}}^2 = L_{\text{post}}\sigma^2$ with parameter-law variance
714 $L_{\text{post}}^2\sigma^2$).

715 **Track 3 — Hellinger backstop (Theorem D.2).** Theorem D.2: $\mathbb{E}\text{Hel}(P_{M_{\tau+}|G}, P_{M_{\tau+}}) \leq$
716 $(1/\sqrt{2})\sqrt{I_M}$ globally, where Hel is between two parameter laws $\mathcal{N}(\mu_1, L_{\text{post}}^2\sigma^2)$ and $\mathcal{N}(\mu_2, L_{\text{post}}^2\sigma^2)$.
717 Hellinger is globally bounded by 1, so the bound is non-trivial when $\sqrt{I_M}/2 < 1$ — i.e., $I_M < 2$
718 nat.

719 **Numerical comparison at balanced scale** ($\tau = \sigma = 1$, $L_{\text{post}} = 1/2$, $L_{\text{post}}\sigma = 1/2$, $L_{\text{post}}^2\sigma^2 = 1/4$,
720 transferred info $I_M = 1$ nat for normalization):

Table 2: Conjugate-Gaussian numerical comparison at balanced scale ($\tau = \sigma = 1$, $I_M = 1$ nat).

Track	Theorem	Bound	Numerical at $I_M = 1$
Track 1	Theorem 4.1	$\mathbb{E}[W_2^2] \leq C_{T_2} I_M = (1/2) I_M$	$\mathbb{E} W_2 \leq 1/\sqrt{2}$
Track 2	Theorem 4.1	$\mathbb{E} d_{FR} \leq \sqrt{2I_M}$	$\mathbb{E} d_{FR} \leq \sqrt{2}$
Track 5.5 Euclidean	Theorem D.4	$\mathbb{E} \ \Delta\mu_+\ \leq L_{\text{post}}\sigma\sqrt{2I_M} = (1/2)\sqrt{2I_M}$	$\mathbb{E} \ \Delta\mu_+\ \leq 1/\sqrt{2}$
Hellinger	Theorem D.2	$\mathbb{E} \text{Hel} \leq \sqrt{I_M}/2$	$\mathbb{E} \text{Hel} \leq 1/\sqrt{2}$

721 Track 1 and Track 5.5 agree exactly at balanced scale ($\mathbb{E} W_2 = \mathbb{E} \|\Delta\mu_+\|_{\text{Eucl}} = 1/\sqrt{2}$ at $I_M = 1$)
722 — they bound the same Euclidean displacement; Track 5.5 is just Track 2 with the Fisher-Rao-to-
723 Euclidean conversion factor $L_{\text{post}}\sigma$ on the parameter chart. The Euclidean translation goes to zero in
724 both extreme limits (vanishing prior-dominant and vanishing likelihood-dominant). At $I_M = 1$ nat
725 the Hellinger bound saturates the global $\sqrt{1/2} \approx 0.71$; for $I_M \ll 1$ Track 1 / Track 5.5 are tighter;
726 for $I_M \gtrsim 2$ the Hellinger backstop is the only non-vacuous bound.

727 D Track 2 companions and parametric Euclidean translations

728 The locally-tight Fisher-Rao bound (Theorem 4.1’s Track 2 instantiation) is one of three Fisher-
729 Rao-spine bounds; the other two — a globally-valid Fisher-Rao backstop and a Hellinger backstop
730 — recover under strictly weaker hypotheses, with universal constants of their own. Together with
731 the conjugate-Gaussian Euclidean translation (review-defusing the no-go’s apparent scale-family
732 pathology) and the exponential-family Euclidean generalization, they form the Track 2 family.

733 D.1 Track 2 globally-valid backstop

734 Outside the uniform-locality regime — when a few rare goals carry disproportionately large trans-
 735 ferred information, or when the operating distribution sits at moderate-to-large I_M — the slice-wise
 736 expansion’s $(1 + o(1))$ remainder is no longer uniformly sharp. A global Fisher-Rao bound is available
 737 under strictly weaker hypotheses, at the cost of a slightly larger constant.

738 **Theorem D.1** (Track 2 global backstop). *Under (H1) and (PI) — for all values of transferred*
 739 *information,*

$$740 \mathbb{E} d_{FR}(P_{M_{\tau+}|e, M_{\tau-}, G}, P_{M_{\tau+}|e, M_{\tau-}}) \leq (\pi/\sqrt{2}) \cdot \sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})}.$$

741 *The constant $\pi/\sqrt{2} \approx 2.22$ is universal, dimension-free, global — no small-information condition*
 742 *required, and (R), (K) are not invoked.*

743 *Proof.* On the unit L^2 -sphere of $\sqrt{p}$, the chord-arc identity (Lemma D.3) gives $\text{Hel}(P, Q) =$
 744 $\sqrt{2} \sin(d_{FR}(P, Q)/4)$ exactly. Lemma D.3 shows $d_{FR}(P, Q) \leq \pi \text{Hel}(P, Q)$ globally, with equal-
 745 ity at the antipode $\text{Hel} = 1$. By Tsybakov²⁹, Lemma 2.4, $2\text{Hel}^2 \leq \text{KL}$ slice-wise. Combin-
 746 ing: $d_{FR}^2 \leq \pi^2 \text{Hel}^2 \leq (\pi^2/2) \text{KL}$ slice-wise. Take $\mathbb{E}_G$, substitute the chain rule Lemma 5.1:
 747 $\mathbb{E} d_{FR}^2 \leq (\pi^2/2) I_M$. Jensen on $\sqrt{x}$: $\mathbb{E} d_{FR} \leq (\pi/\sqrt{2}) \sqrt{I_M}$. $\square$

748 The $\pi/2 \approx 1.57$ overhead vs. the locally-tight $\sqrt{2}$ is exactly the worst-case arc-chord ratio at the unit
 749 L^2 -sphere’s antipode — a geometric maximum, not a slack. Equality is achieved in the limit $\text{Hel} \rightarrow 1$
 750 at the antipode, when goals can drive the post-update measure to a near-disjoint support from the
 751 marginal.

752 D.2 Hellinger backstop

753 The chord companion to Theorem D.1’s arc bound, with the cleanest universal constant.

754 **Theorem D.2** (Hellinger backstop). *Under (H1) and (PI) — for all values of transferred information,*

$$755 \mathbb{E} \text{Hel}(P_{M_{\tau+}|e, M_{\tau-}, G}, P_{M_{\tau+}|e, M_{\tau-}}) \leq (1/\sqrt{2}) \cdot \sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})}.$$

756 *Under (H_{κ}) , the architectural factorization recovers: $\mathbb{E} \text{Hel} \leq (1/\sqrt{2}) \sqrt{\kappa_{\text{processing}} \cdot I(G; \Omega | e, M)}$.*

757 *Proof.* By Tsybakov²⁹, Lemma 2.4, $2\text{Hel}^2(P, Q) \leq \text{KL}(P||Q)$ globally for all probability measures.
 758 Apply slice-wise at each $G = g$, take expectation over G , and substitute the chain rule Lemma 5.1.
 759 $\square$

760 Hellinger satisfies (PI) — it is an f -divergence with $f(t) = (\sqrt{t} - 1)^2/2$ preserved under Markov
 761 morphisms (Csiszár 1967; Liese-Vajda 1987) — and is locally proportional to Fisher-Rao on the
 762 unit-sphere representation. Throughout we adopt the *standard statistician’s* Hellinger convention
 763 $\text{Hel}^2(P, Q) := \frac{1}{2} \|\sqrt{p} - \sqrt{q}\|_{L^2}^2 = 1 - \int \sqrt{pq}$, so $\text{Hel} \in [0, 1]$, and the *Amari-Nagaoka* Fisher-Rao
 764 convention $d_{FR}(P, Q) = 2 \arccos \int \sqrt{pq}$ (twice the great-circle arc on the unit sphere of $\sqrt{p}$ in L^2 ,
 765 $d_{FR} \in [0, \pi]$). Different conventions in the literature differ by factors of 2; the forms above are the
 766 ones consistent with $\text{Hel} \in [0, 1]$, $d_{FR} \in [0, \pi]$, and the $2\text{Hel}^2 \leq \text{KL}$ inequality used in the proof.

767 The three Fisher-Rao-spine bounds — locally-tight $\sqrt{2}$ (under (H1)+(H4’)+(PI)+(R)+(K)), globally-
 768 valid $\pi/\sqrt{2}$ (Theorem D.1, under (H1)+(PI) only), and Hellinger chord $1/\sqrt{2}$ (Theorem D.2, under
 769 (H1)+(PI) only) — share the same intrinsic geometry and have strictly nested hypotheses. There
 770 is no “metric switch” between regimes: macroscopic hallucination is in Fisher-Rao directly via
 771 Theorem D.1; Theorem D.2 is the chord companion on the same unit L^2 -sphere.

772 D.3 Hellinger-Fisher-Rao chord-arc geometry

773 **Lemma D.3** (Global chord-arc bound on the unit L^2 -sphere). *For any probability measures P, Q ,*
 774 $d_{FR}(P, Q) \leq \pi \cdot \text{Hel}(P, Q)$, *with equality at the antipode $\text{Hel}(P, Q) = 1$ (correspondingly*
 775 $d_{FR}(P, Q) = \pi$).

776 *Proof.* From $d_{FR} = 4 \arcsin(\text{Hel}/\sqrt{2})$, define $\phi : (0, 1] \rightarrow (0, \pi]$ by $\phi(h) := 4 \arcsin(h/\sqrt{2})/h$,
777 so $d_{FR}/\text{Hel} = \phi(\text{Hel})$. Set $g(h) := 4 \arcsin(h/\sqrt{2})$. Differentiating: $g'(h) = 4/\sqrt{2-h^2}$ for
778 $h \in [0, \sqrt{2})$, $g''(h) = 4h/(2-h^2)^{3/2} > 0$ on $(0, \sqrt{2})$ — so g is *strictly convex* on this interval, with
779 $g(0) = 0$. By the secant-slope monotonicity for convex functions vanishing at zero, $\phi(h) = g(h)/h$
780 is monotonically increasing on $(0, \sqrt{2})$, with limits $\lim_{h \rightarrow 0^+} \phi(h) = g'(0) = 4/\sqrt{2} = 2\sqrt{2}$ and
781 $\phi(1) = g(1) = 4 \arcsin(1/\sqrt{2}) = \pi$. Hence $\phi(h) \leq \pi$ for all $h \in (0, 1]$, with equality at $h = 1$.
782 $\square$

783 **Local proportionality** As $d_{FR} \rightarrow 0$, $\text{Hel} \approx d_{FR}/(2\sqrt{2})$, so Theorem 4.1’s Track 2 locally-tight
784 bound $\mathbb{E} d_{FR} \leq \sqrt{2I_M}$ jointly with Lemma D.3 gives $\mathbb{E} \text{Hel} \lesssim (1/2)\sqrt{I_M}$ in the small-information
785 limit — *tighter* than Theorem D.2’s global $(1/\sqrt{2})\sqrt{I_M}$ by a factor of $\sqrt{2}$. So in the (H4’) regime,
786 Theorem 4.1 + Lemma D.3 jointly give the tightest Hellinger bound; Theorem D.2 (under (PI) only,
787 no (H4’)) is the global statement.

788 D.4 Conjugate-Gaussian Euclidean translation

789 When (H1) is realized by a parametric family with known Fisher information, the locally-tight Fisher-
790 Rao bound translates to an explicit Euclidean bound on parameter displacement. The conjugate-
791 Gaussian case is paradigmatic and review-defusing: *Euclidean displacement does not blow up* in the
792 prior-dominant limit, contrary to a naive reading of the no-go.

793 **Theorem D.4** (Conjugate-Gaussian Euclidean bound). *Under the conjugate-Gaussian setup ($\theta \sim$
794 $\mathcal{N}(0, \tau^2)$, $\Omega \mid \theta \sim \mathcal{N}(\theta, \sigma^2)$, goal-conditional likelihood-mean shift $\Omega \mid G \sim \mathcal{N}(\beta(G), \sigma^2)$),
795 the post-update parameter law is the Gaussian $\mu_+ \mid G \sim \mathcal{N}(L_{\text{post}}\beta(G), L_{\text{post}}^2\sigma^2)$ with $L_{\text{post}} =$
796 $\tau^2/(\sigma^2 + \tau^2)$. The locally-tight Fisher-Rao bound translates to the Euclidean bound on random-
797 parameter displacement*

$$798 \mathbb{E} |\Delta\mu_+|_{\text{Eucl}} \leq L_{\text{post}} \sigma \cdot \sqrt{2I_M} \cdot (1 + o(1)),$$

799 with prefactor $L_{\text{post}} \sigma = \tau^2 \sigma / (\sigma^2 + \tau^2) \leq \tau/2$ uniformly. Under (H_κ) , $\mathbb{E} |\Delta\mu_+|_{\text{Eucl}} \leq$
800 $L_{\text{post}} \sigma \sqrt{2\kappa_{\text{processing}}} I(G; \Omega \mid e, M) (1 + o(1))$.

801 *Proof.* The post-update parameter law $\mu_+ \mid G$ is Gaussian with mean $L_{\text{post}}\beta(G)$ and variance $L_{\text{post}}^2\sigma^2$
802 (variance preserved across goals because σ_{post}^2 is fixed by the conjugate-update structure). The Fisher
803 metric on the manifold $\{\mathcal{N}(\mu, L_{\text{post}}^2\sigma^2) : \mu \in \mathbb{R}\}$ in the μ -coordinate is $\mathbf{I}(\mu) = 1/(L_{\text{post}}^2\sigma^2)$, so the
804 Fisher-Rao geodesic distance between two such laws is $d_{FR}(\mu_1, \mu_2) = |\mu_1 - \mu_2|/(L_{\text{post}}\sigma)$ — a flat
805 metric pulled back from $\mathbb{R}$. Hence $|\Delta\mu_+|_{\text{Eucl}} = L_{\text{post}}\sigma \cdot d_{FR}$. Take expectations and substitute the
806 locally-tight Fisher-Rao bound. $\square$

807 The prefactor $L_{\text{post}}\sigma = \tau^2\sigma/(\sigma^2 + \tau^2)$ is bounded above by $\tau/2$ uniformly (AM-GM at the maximizer
808 $\sigma = \tau$) and *vanishes in both extreme limits*: $L_{\text{post}}\sigma \rightarrow 0$ as $\sigma \rightarrow \infty$ (very noisy observations: the
809 architecture learns essentially nothing from Ω regardless of goal) and $L_{\text{post}}\sigma \rightarrow 0$ as $\sigma \rightarrow 0$ (very
810 sharp observations: $L_{\text{post}} \rightarrow 1$ but $\sigma \rightarrow 0$, so even though the posterior tracks data tightly the data
811 variance vanishes). Maximum at balanced $\sigma = \tau$. The naive “ $\sigma\sqrt{2I}$ ” reading is wrong (correct
812 prefactor is $L_{\text{post}}\sigma$); the “ $\sigma_{\text{post}}\sqrt{2I}$ ” reading is *also* wrong (it conflates inside-posterior covariance
813 $\sigma_{\text{post}}^2 = L_{\text{post}}\sigma^2$ with parameter-law variance $L_{\text{post}}^2\sigma^2$). The Section 4.1 no-go still applies as stated to
814 *charts*: $L_{\text{post}}\sigma$ is family-specific.

815 D.5 Exponential-family Euclidean generalization

816 For an exponential family $P_\theta(x) = h(x) \exp(\theta^\top T(x) - A(\theta))$ with natural parameter $\theta \in \Theta \subseteq \mathbb{R}^d$
817 and log-partition A , the Fisher information metric is $\mathbf{I}(\theta) = \nabla^2 A(\theta)$. The infinitesimal Fisher-Rao
818 expansion gives $d_{FR}^2(P_{\theta_1}, P_{\theta_2}) = (\theta_1 - \theta_2)^\top \mathbf{I}(\theta^*)(\theta_1 - \theta_2) + O(\|\theta_1 - \theta_2\|^3)$ for θ^* on the geodesic
819 between θ_1, θ_2 . So Euclidean displacement on the natural parameter satisfies $\|\theta_1 - \theta_2\|^2 \leq \mathbf{I}_{\min}^{-1} \cdot d_{FR}^2$
820 where $\mathbf{I}_{\min}$ is the minimum eigenvalue of $\mathbf{I}$ along the geodesic.

821 **Theorem D.5** (Exponential-family Euclidean bound). *For an exponential family in natural-parameter*
822 *coordinates with Fisher-information minimum-eigenvalue $\mathbf{I}_{\min}$ along the goal-induced geodesic, the*
823 *locally-tight Fisher-Rao bound translates to*

$$\mathbb{E} \|\Delta\theta_{\text{bias}}\|_{\text{Eucl}} \leq \mathbf{I}_{\min}^{-1/2} \cdot \sqrt{2I_M} \cdot (1 + o(1)),$$

$$\text{and under } (H_\kappa): \mathbb{E} \|\Delta\theta_{\text{bias}}\|_{\text{Eucl}} \leq \mathbf{I}_{\min}^{-1/2} \sqrt{2\kappa_{\text{processing}} I(G; \Omega \mid e, M)} (1 + o(1)).$$

Proof. By Theorem 4.1’s Track 2 instantiation, $\mathbb{E} d_{FR}^2 \leq 2I_M(1 + o(1))$. By the infinitesimal Fisher-Rao expansion, $\|\theta_1 - \theta_2\|^2 \leq \mathbf{I}_{\min}^{-1} d_{FR}^2$. Take expectations and Jensen. $\square$

Worked examples *Bernoulli on logit chart, restricted to $p \in [\delta, 1 - \delta]$:* $\mathbf{I}(\theta) = p(1 - p)$ has no positive lower bound on the full chart $\theta \in \mathbb{R}$; restricting to $p \in [\delta, 1 - \delta]$ gives $\mathbf{I}_{\min} \geq \delta(1 - \delta)$ and finite, dimension-free Euclidean bound $\mathbb{E} \|\Delta\theta\|_{\text{Eucl}} \leq [\delta(1 - \delta)]^{-1/2} \sqrt{2I_M}$ — contingent on boundary avoidance. *Multinomial on log-ratio chart:* Fisher minimum eigenvalue bounded below by $\min_i p_i$ on the simplex interior; for δ -bounded models, $\mathbb{E} \|\Delta\theta\|_{\text{Eucl}} \leq \delta^{-1/2} \sqrt{2I_M}$. *Gaussian-mean chart with fixed variance v :* $\mathbf{I}(\mu) = 1/v$ constant on the manifold; $\mathbb{E} \|\Delta\mu\|_{\text{Eucl}} \leq \sqrt{v} \sqrt{2I_M}$. *Gaussian-scale chart $\theta = \sigma$:* $\mathbf{I}(\sigma) = 2/\sigma^2 = \mathbf{I}_{\min}^{-1/2} = \sigma/\sqrt{2}$ grows with σ , recovering the no-go’s pathology consistent with Section 4.1; reparameterizing to $\theta = \log \sigma$ gives $\mathbf{I}(\log \sigma) = 2$ constant, restoring a finite Euclidean bound.

The pattern: for natural-parameter charts on simplex / discrete-outcome models with the goal-induced geodesic restricted away from the boundary, the Euclidean Track 2 translation is dimension-free and finite; for scale-family parametrizations off the natural-parameter chart, the prefactor grows in σ or λ^{-1} — recovering Section 4.1’s pathology consistent with the no-go.

E Proofs of main results

This appendix collects full proofs deferred from §4–§5 of the main text. Proof structure: each section restates the formal claim from main text, then provides the full derivation. Auxiliary lemmas used in proofs are stated where they first arise.

E.1 Track 1 cascade — full derivation

The umbrella theorem’s Track 1 instantiation (Theorem 4.1 under (H1) + (H2’)) follows from a two-step composition: chain rule on the post-update law, then slice-wise Talagrand T_2 .

Proof. Step 1 — Chain rule of relative entropy on the post-update law. By Cover and Thomas 2006^[25] Theorem 2.5.3 applied to the conditional law of $M_{\tau+}$ given $(e_\tau, M_{\tau-})$ marginalized over G :

$$\mathbb{E}_G [\text{KL}(P_{M_{\tau+}|e, M_{\tau-}, G} \parallel P_{M_{\tau+}|e, M_{\tau-}})] = I(G; M_{\tau+} \mid e_\tau, M_{\tau-}).$$

The (H1) standard-Borel regularity makes this hold in the abstract-spaces form (Polyanskiy and Wu²⁰, Theorem 3.4; Gray²¹, Theorem 5.4). The right-hand side is *transferred* goal-information into the post-update model state.

Step 2 — Slice-wise Talagrand T_2 . Apply Hypothesis 4.2 slice-wise at each $G = g$:

$$W_2^2(P_{M_{\tau+}|e, M, G=g}, P_{M_{\tau+}|e, M}) \leq C_{T_2} \cdot \text{KL}(P_{M_{\tau+}|e, M, G=g} \parallel P_{M_{\tau+}|e, M}).$$

Take expectation over G and substitute Step 1:

$$\mathbb{E} [W_2^2(P_{M_{\tau+}|G}, P_{M_{\tau+}})] \leq C_{T_2} \cdot I(G; M_{\tau+} \mid e_\tau, M_{\tau-}).$$

Jensen’s inequality on $\sqrt{x}$ delivers the distance form

$$\mathbb{E} W_2(P_{M_{\tau+}|G}, P_{M_{\tau+}}) \leq \sqrt{C_{T_2}} \cdot \sqrt{I(G; M_{\tau+} \mid e_\tau, M_{\tau-})}. \quad \square$$

The architectural-corollary form (Corollary 4.7 under (H $_\kappa$)) follows by substituting $I(G; M_{\tau+} \mid e, M_{\tau-}) \leq \kappa_{\text{processing}} \cdot I(G; \Omega_\tau \mid e, M_{\tau-})$ on the right-hand side; the substitution is *definitional* per (4), with the boundedness of $\kappa_{\text{processing}}$ being the substantive content of (H $_\kappa$).

864 E.2 Track 2 cascade — full derivation

865 The umbrella theorem’s Track 2 instantiation (under (H1) + (H4’) + (PI) + (R) + (K)) follows in one
866 step using the KL-to-Fisher-Rao expansion.

867 *Proof.* Apply Lemma 5.2 under (H1) + (H4’) slice-wise at each $G = g$ to get

$$868 d_{FR}^2(P_{M_{\tau+|e,M,G=g}}, P_{M_{\tau+|e,M}}) \leq 2 \text{KL}(P_{M_{\tau+|G=g}} \| P_{M_{\tau+}}) (1 + o(1)).$$

869 Under (H4’) the remainder is controlled uniformly: with bounded Amari-Chentsov tensor
870 $|T_{ijk}\delta^i\delta^j\delta^k| \leq \mathcal{T}_\star \|\delta\|^3$ on the goal-induced neighborhood and minimum Fisher eigenvalue $\mathbf{I}_{\min} > 0$,
871 the slice-wise inequality $d_{FR}^2(P_{M_{\tau+|G=g}}, P_{M_{\tau+}}) \leq 2 \text{KL}_g(1 + R_3(\delta_\star))$ holds with $R_3(\delta_\star) =$
872 $(\mathcal{T}_\star/3)\sqrt{2\delta_\star/\mathbf{I}_{\min}^3} \rightarrow 0$ as $\delta_\star \rightarrow 0$.

873 Take expectation over G and substitute the chain-rule identity from Lemma 5.1:

$$874 \mathbb{E}[d_{FR}^2(P_{M_{\tau+|G}}, P_{M_{\tau+}})] \leq 2 \cdot I(G; M_{\tau+} | e_\tau, M_{\tau-}) \cdot (1 + o(1)).$$

875 Jensen on $\sqrt{x}$ delivers the distance form

$$876 \mathbb{E} d_{FR}(P_{M_{\tau+|G}}, P_{M_{\tau+}}) \leq \sqrt{2} \cdot \sqrt{I(G; M_{\tau+} | e_\tau, M_{\tau-})} \cdot (1 + o(1)). \quad \square$$

877 The constant $\sqrt{2}$ emerges from the second-order coefficient $\frac{1}{2}$ in the KL-to-Fisher expansion (Amari
878 and Nagaoka¹⁸ §3.7 Theorem 3.1) — exactly what (K) crystallizes as a normalization. Outside (H4’),
879 the locally-tight bound’s $(1 + o(1))$ remainder is no longer uniformly sharp; companion bounds in
880 Section D cover the moderate-to-large I_M regime under strictly weaker hypotheses (Track 2 global,
881 Hellinger backstop).

882 E.3 No-go on Euclidean chart norms — full proof

883 Theorem 4.4’s proof uses a chart-rescaling lemma plus a one-line consequence.

884 **Lemma E.1** (Chart-rescaling sensitivity of Euclidean displacement). *Let $\mathcal{M}$ be a statistical manifold*
885 *and $\phi : U \rightarrow \mathbb{R}^d$ a chart on an open set $U \subseteq \mathcal{M}$. For any $a > 0$, write $\phi_a := a\phi$ for the rescaled*
886 *chart. For every pair of probability laws μ, ν on U (in particular, the goal-conditional and goal-*
887 *marginal post-update laws of (3)): (a) The chart-Euclidean Wasserstein distance scales linearly:*
888 *$W_2^{\phi_a}(\mu, \nu) = a \cdot W_2^\phi(\mu, \nu)$. (b) Conditional KL, conditional mutual information, Fisher-Rao geodesic*
889 *distance, and Hellinger distance are all chart-invariant: they take the same value under ϕ and ϕ_a .*

890 *Proof.* (a) The pushforward under the linear map $x \mapsto ax$ rescales Euclidean distances by a , hence
891 rescales any optimal coupling’s transport cost. (b) KL depends only on the underlying densities
892 and dominating measure (Radon-Nikodym derivatives are reparameterization-invariant). The Fisher
893 metric tensor transforms covariantly under chart change, so the integrated geodesic distance is invariant.
894 Hellinger as an f -divergence is reparameterization-invariant. $\square$

895 *Proof. Proof of Theorem 4.4.* Suppose for contradiction that a coordinate-independent $C_0 < \infty$
896 exists such that for every chart ϕ on $\mathcal{M}$ and every architecture satisfying (H1)+(H2’) with non-zero
897 post-update displacement,

$$898 \|\Delta M_{\text{bias}}\|_{\text{Eucl}, \phi} \leq C_0 \sqrt{I(G; M_{\tau+} | e_\tau, M_{\tau-})}.$$

899 For any architecture realizing $W_2^\phi(P_{M_{\tau+|G,e,M_{\tau-}}}, P_{M_{\tau+|e,M_{\tau-}}}) > 0$ and finite positive transferred
900 information, applying the supposed bound under the rescaled chart ϕ_a gives, by Lemma E.1,

$$901 a \cdot \|\Delta M_{\text{bias}}\|_{\text{Eucl}, \phi} = \|\Delta M_{\text{bias}}\|_{\text{Eucl}, \phi_a} \leq C_0 \sqrt{I},$$

902 with the right-hand side unchanged across charts. Taking $a \rightarrow \infty$ contradicts the fixed $C_0 \sqrt{I}$. The
903 same conclusion holds for the architectural-corollary form $C \sqrt{\kappa_{\text{processing}}} \cdot I(G; \Omega | e, M)$ under $(\mathbf{H}_\kappa)$:
904 both sides inherit Lemma E.1’s chart-(non)invariance, and the contradiction goes through verbatim.
905 $\square$

The proof avoids type-shifts between Dirac points and pushforward laws, never invokes (H2')-violating two-point constructions, and does not assert any specific value of I — it only uses that some architecture realizes finite positive W_2^ϕ and finite positive I , automatic on any non-trivial goal-coupled architecture (e.g., the conjugate-Gaussian setup with any $\sigma, \tau > 0$ and a non-degenerate continuous goal distribution). The Gaussian scale family with two charts (Chart A: σ ; Chart B: $\log \sigma$) is the canonical illustration: same intrinsic geometry, Chart-A Euclidean displacement grows linearly while Chart-B stays bounded as the operating scale grows.

913 E.4 Čencov uniqueness and sharpness — full proof

914 Theorem 4.5 has two parts.

915 *Proof. Proof of (a) — Uniqueness of the metric.* (PI) + (R) invokes Čencov’s uniqueness theorem^[17]
 916 (modern treatment Ay et al.³⁰, Theorem 5.1): on $\mathcal{M}$, the Fisher information metric $\mathbf{I}$ is the unique
 917 (up to a global positive scalar $c > 0$) Riemannian metric invariant under Markov morphisms. Write
 918 $d_{\mathcal{M}} = \sqrt{c} d_{FR}$ where d_{FR} is at the Amari-Nagaoka normalization $\mathbf{I}(\theta) = -\mathbb{E}[\nabla^2 \log p_\theta]$. The KL-
 919 to-Fisher second-order expansion (Amari and Nagaoka¹⁸ §3.7 Theorem 3.1) gives $\text{KL}(P_\theta \| P_{\theta+\delta}) =$
 920 $\frac{1}{2} d_{FR}^2 + O(d_{FR}^3)$, so $\frac{1}{2} d_{\mathcal{M}}^2 = (c/2) d_{FR}^2 + O(d_{FR}^3)$. Constraint (K) requires $\frac{1}{2} d_{\mathcal{M}}^2$ agree with KL at
 921 second order, forcing $c = 1$ and $d_{\mathcal{M}} = d_{FR}$. $\square$

922 *Proof. Proof of (b) — Sharpness of the constant.* The conjugate-Gaussian Class 1 (Sepa-
 923 rated) family of Section C with $\text{Var}_G(\beta) \rightarrow 0$ and $I_M = \text{Var}_G(\beta)/(2\sigma^2)$ held fixed has
 924 $\mathbb{E}[d_{FR}^2(P_{M_{\tau+}|G}, P_{M_{\tau+}})] = \text{Var}_G(\beta)/\sigma^2 = 2I_M$ exactly (verified in Section C). Hence
 925 $\sup\{\mathbb{E} d_{FR}^2/I_M : \text{instances satisfying (H1) + (PI) + (R) + (K) + (H4')\} \geq 2$, so any candidate con-
 926 stant $C^2 < 2$ fails on this family. Combined with (a), $C = \sqrt{2}$ is the unique sharp upper-bound
 927 constant. $\square$

928 The upshot: once we adopt (PI) at full Markov-morphism strength, plus Riemannian structure, plus KL
 929 as the cascade’s information coordinate, the *only* such-invariant choice of norm on $\mathcal{M}$ is Fisher-Rao,
 930 and the *unique sharp* upper-bound constant is $\sqrt{2}$. Alternative chart-independent commitments —
 931 TV with Pinsker, Hellinger-as-divergence, χ^2 , Rényi — give weaker bounds: TV-Pinsker is non-tight
 932 (the constant is universal but the inequality is loose except at coincident measures); Hellinger-as-
 933 divergence and χ^2 -as-divergence violate (R); switching divergences on the RHS violates (K). The
 934 global Theorem D.1’s $\pi/\sqrt{2}$ is also sharp in its own (PI)-only regime — achieved in the limit $\text{Hel} \rightarrow 1$
 935 at the unit L^2 -sphere’s antipode.

936 E.5 Coupled-class attention connectivity — full proof

937 *Proof. Proof of Lemma 3.5.* By induction on layer depth.

938 *Layer 0 (embedding).* $h_0^{(i)}$ depends only on X_i — the position- i token embedding. Goal positions
 939 i_G produce embeddings $h_0^{(i_G)}$ that are functions only of the goal tokens at those positions; same for
 940 evidence positions i_E . No cross-position dependence yet.

941 *Layer $\ell \geq 1$.* A standard transformer block (pre-LN or post-LN) computes

$$942 h_\ell^{(j)} = h_{\ell-1}^{(j)} + \text{Attn}_\ell(\text{LN}(h_{\ell-1}))^{(j)} + \text{MLP}_\ell(\text{LN}(h_{\ell-1}^{(j)})),$$

943 where LayerNorm and the MLP are *position-wise* (the operation at position i depends only on $h_{\ell-1}^{(i)}$,
 944 not on other positions). This induces a directed position-graph with two edge types at each layer: (i) a
 945 *same-position* residual edge $h_{\ell-1}^{(j)} \rightarrow h_\ell^{(j)}$ always present (via the residual stream); (ii) *cross-position*
 946 attention edges $h_{\ell-1}^{(i)} \rightarrow h_\ell^{(j)}$ present whenever $\text{Attn}_\ell^{(j,i)}(\theta) \neq 0$. Position-wise LayerNorm and MLP
 947 preserve both edge types.

948 *Inductive claim.* For every $\ell \geq 0$ and every input position $i \leq j$, $h_\ell^{(j)}$ has a directed-graph path back
 949 to position i in the input.

950 *Inductive step.* The same-position residual chain gives $i = j$. For $i \neq j$ with $i \leq j$, non-degenerate
 951 attention at any single layer $\ell' \leq \ell$ along the path gives a cross-position edge $h_{\ell'-1}^{(i)} \rightarrow h_{\ell'}^{(j)}$, which

composes with the residual chain on either side to yield a directed-graph path from input position i to $h_\ell^{(j)}$.

Coupled-class conclusion. For any goal position $i_G \leq j$, $h_\ell^{(j)}$ has a path back to position i_G in the input whenever attention is non-degenerate on the (j, i_G) edge at some layer $\ell' \leq \ell$. So under non-degenerate attention, G (encoded at positions i_G) is causally upstream of every quantity contributing to a post-update model state read off from position $j \geq \max(i_G \cup i_E)$. Class 3 (Coupled) by construction.

Robustness to architectural variants. - RMSNorm, GroupNorm, FlashAttention preserve the position-graph: position-wise normalization in the first two cases; mathematically-equivalent attention implementation in the third. - Causal masking preserves $\text{Attn}_\ell^{(j, i_G)} \neq 0$ for $i_G \leq j$ — the regime the lemma covers. - Sliding-window or sparse attention preserves connectivity by composition across layers when the window/sparsity pattern admits a multi-hop path from i_G to j , which is the case for trained LLMs in their operating regime. - Architectures with explicit goal-to-output attention masking, or token-dropping at intermediate layers, fall outside the lemma’s scope and are appropriately Class 2 (Partial). $\square$

Lemma 3.5 establishes a *downstream-output graph-reachability* claim: at every layer and every position causally downstream of the goal and evidence under standard causal masking, the activation has a directed-graph path back to at least one goal position. *Three caveats:*

- *Graph reachability is not the same as causal-influence magnitude.* A directed-graph path with near-zero attention weights produces a formal edge with negligible quantitative influence on $h_\ell^{(j)}$. The connectivity claim is therefore *structural* — about the existence of a path, not the magnitude of effect — and translates into Class 3 (Coupled) classification under the Markov-blanket conditional-independence reading. Bounding $\kappa_{\text{processing}}$ requires the architectural-bandwidth conditions of Section B.3 (information-bottleneck on the goal channel; bounded direct-channel capacity uniformly over operating distributions).
- *Causal masking restricts the lemma’s scope to downstream positions.* For $j < \min(i_G)$ — positions strictly upstream of any goal token — there is no causal path from goal to $h_\ell^{(j)}$ under causal attention masks; those positions are Class 1 (Separated) in their own right.
- *Sparse and sliding-window attention require a multi-hop reachability assumption.* The connectivity proof composes single-layer attention edges across layers; for sliding-window or sparse-attention architectures, multi-hop connectivity from i_G to j requires at least one composition of single-layer paths through the window/sparsity pattern. Architectures whose sparsity pattern excludes goal-to-output communication entirely fall outside the lemma’s hypothesis and are Class 2 (Partial).

The bias-bound *interpretation* — bounding the displacement of an implicit posterior under goal-conditional re-prompting — additionally invokes the in-context-learning correspondence^[12–15] between the next-token distribution and an implicit posterior. The structural Coupled-class claim of Lemma 3.5 is robust without it; the bias-bound *application* is conditional on the in-context-learning correspondence, which has its own scope (in-distribution prompts: tight; OOD/jailbreak: degraded to a bound on output-distribution displacement rather than belief-state displacement).

F Extended related work

§2 Related Work names the two main lineages this paper composes (LLM hallucination theory; Bayesian inverse-problems posterior stability). This appendix covers the additional adjacent strands — Fisher-Rao Bayesian sensitivity, Owahdi-Scovel-Sullivan brittleness, mean-field variational inference stability, and the Markov-blanket apparatus from active inference — that sit close enough to require explicit cite-and-distinguish but do not resolve the goal-coupling-displacement gap.

F.1 Strand 1: LLM hallucination theory

The recent statistical-inevitability and capacity-impossibility lineage targets the *frequency* of hallucination on architecture-agnostic grounds. Kalai and Vempala 2023^[1] establish that calibrated language

models must hallucinate at a rate close to the fraction of singleton facts in the training data — a statistical lower bound that holds independently of the transformer architecture. Kalai et al. 2026^[2] extend this to argue that hallucinations originate as binary-classification errors compounded by training-and-evaluation incentives that reward guessing under uncertainty. Karbasi et al. 2025^[3] reduce hallucination *detection* to language identification in the Gold-Angluin framework, showing that detection is impossible from positive examples alone but feasible with expert-labeled negatives. Wu et al. 2024^[4] establish a VC-dimension-based no-free-lunch result for non-hallucinating learning. Suzuki et al. 2025^[5] give a probabilistic positive result — hallucinations can be made statistically negligible — counterbalancing the computability-theoretic inevitability arguments. Guo and Li 2026^[31] frame hallucination as a rate-distortion theorem for membership testing under capacity constraints. Liu et al. 2025^[32] formalize hallucinations as estimation failures with high-probability lower bounds on the hallucinate rate. Chlon et al. 2025^[33] frame transformers as Bayesian-in-expectation-not-realization with $O(\log n)$ permutation dispersion. Zeng and al 2026^[34] decompose hallucination risk into data-driven and reasoning-driven components.

Our distance from this strand: we bound a *displacement*, not a frequency, under named *geometric* assumptions, and we engage the *architecture’s coupling structure* directly via $\kappa_{\text{processing}}$. The bound is not a substitute for the frequency bounds; it is an orthogonal axis. Among the recent papers, Zeng et al.’s decomposition is closest in shape to our $\kappa \times I$ factorization but differs on structural axis (training-time vs inference-time, vs our architectural-vs-informational).

1019 F.2 Strand 2: Bayesian inverse-problems posterior stability

The transport-inequality and Lipschitz-posterior machinery driving Track 1 is mature. Stuart 2010^[6] establishes well-posedness of Bayesian inverse problems on function spaces, with Lipschitz continuity of the posterior in the data. Sprungk 2020^[7] extends this to local Lipschitz stability with respect to perturbations of the prior and the log-likelihood, in TV / Hellinger / Wasserstein / KL distances. Cvetković and Lie 2025^[8] match the upper bounds with lower bounds, emphasizing that posterior sensitivity *increases* as the posterior concentrates. Dolera and Mainini 2023^[9] obtain Lipschitz-continuity of probability kernels in the optimal-transport framework, with explicit constants in terms of Fisher-information functionals and weighted Poincaré constants — the closest-in-spirit prior result to Track 1’s form. Garbuno-Iñigo et al. 2023^[10] develop integral-probability-metric perturbation analysis under log-Sobolev-style functional inequalities. Latz 2020^[35] provides well-posedness under weaker continuity. [? lie-sullivan-teckentrup-2017] give Hellinger-distance bounds for randomized forward models. Hosseini et al. 2025^[11] develop conditional optimal transport on function spaces for amortized Bayesian inference, with explicit regularity estimates on conditioning maps from prior to posterior — the closest single-paper match for Track 1’s cascade. Dolera et al. 2024^[36] derive Wasserstein-based posterior contraction rates. Del Moral 2026^[37] gives Sinkhorn / Schrödinger bridge contraction via log-Sobolev — most recent transport-inequality-plus-LSI work, adjacent to Track 1’s machinery.

Our distance from this strand: we apply this machinery to *belief-goal-coupled* inference under an architectural classification. None of these papers address goal-conditional bias under explicit architectural-coupling factorization. The closest, Hosseini-Hsu-Taghvaei, is amortized Bayesian inference rather than goal-coupling bias; the cascade overlaps but the application differs.

1041 F.3 Strand 3: Fisher-Rao geometry for Bayesian sensitivity

Kurtek and Bharath 2015^[38] provide a Fisher-Rao framework for Bayesian sensitivity analysis under ε -contamination classes — the closest existing Fisher-Rao deployment to ours. Same geometric machinery; different target (sensitivity vs universal-constant bias bound). Lebanon 2004^[39] extends Čencov-Campbell uniqueness to conditional information geometry — load-bearing precursor for the conditional-distribution setting in which our bound lives. Čencov 1982^[17] is the original uniqueness result for the Fisher metric under invariance under sufficient statistics; Amari and Nagaoka 2000^[18] give the modern treatment of the KL-to-Fisher second-order expansion. McCulloch 1989^[40] and Ruggeri and Wasserman 1993^[41] develop earlier Fisher-information-flavored sensitivity analyses; both are foundational but produce neither a universal constant nor a goal-coupling bound.

1051 **F.4 Strand 4: Bayesian brittleness (handle carefully)**

1052 Owahdi et al., Owahdi et al. 2015, 2015^[23,24] establish that Bayesian inference is brittle under finite
1053 information in continuous settings: arbitrarily close models on the same data can yield arbitrarily
1054 different posterior conclusions, and any prior + model can be perturbed slightly to achieve any desired
1055 posterior. Owahdi and Scovel 2017^[42] develops the qualitative-robustness companion analysis.

1056 A reader might worry that the brittleness results contradict our Track 1 stability bound. They do not.
1057 Owahdi-Scovel-Sullivan’s no-go is for *arbitrary stability under finite information* in fixed metrics
1058 (TV, generalized moments) — they construct adversarial perturbations *outside* the well-posed regime.
1059 Our Theorem 4.1 (Track 1 instantiation) carries explicit (H1) + (H2′) hypotheses that exclude their
1060 brittleness regime: (H2′)’s post-update concentration is precisely the regularity that brittleness requires
1061 *failing*. The two no-gos constrain different things; our Theorem 4.4 is for *universal-constant absent*
1062 *the (PI) + (R) + (K) triple*, which is a third, distinct statement. The honest reading: three different
1063 no-go results, three different scope conditions, none implying or contradicting the others.

1064 **F.5 Strand 5: Adjacent recent work**

1065 Sheng et al. 2025^[43] establish stability of mean-field variational inference under strongly log-concave
1066 targets — adjacent to Track 1’s transport-inequality machinery. Cattiaux and Guillin 2021^[44] develop
1067 functional inequalities (Poincaré, Cheeger, log-Sobolev) for perturbed measures with explicit constants
1068 — background machinery for Hypothesis 4.2. Ley et al. 2015^[45] give Wasserstein-distance bounds for
1069 nested densities via the Stein method. Cvetković et al. 2023^[46] address observation-operator choice
1070 for misspecified inverse problems. Zahm et al. 2018^[47] provide certified dimension reduction for
1071 nonlinear Bayesian inverse problems. Kleijn and van der Vaart 2006^[48] study misspecified Bayesian
1072 inference asymptotics; Shalizi 2009^[49] studies dynamics of Bayesian updating under dependent
1073 data and misspecified models. Seo et al. 2026^[50] derive W_2 -stability bounds for tangent-linearized
1074 Gaussian inference on smooth manifolds — adjacent for the manifold-Bayesian-inference framing.

1075 In LLM-stability-flavored work: Xu 2025^[51] (*Policy Cliff*) analyzes RLHF stability for LLMs/LRMs
1076 through reward-policy maps; Su-Kempe-Ullrich 2024 address jailbreaking from a statistical per-
1077 spective; Biau and Boyer 2026^[52] develop statistical proxies for retrieval-augmented generation;
1078 Hyeon et al. 2026^[53] address action-sufficient goal representations in RL. None of these target the
1079 goal-coupling bias quantity (3) under an architectural-coupling factorization.

1080 **F.6 Strand 6: Active inference and Markov-blanket apparatus**

1081 The Markov-blanket apparatus from active inference^[54–56] supplies the conditional-independence
1082 machinery our architectural-separation classification invokes. We adopt the Pearl-blanket reading^[16]
1083 — the technical conditional-independence statement — and explicitly do not adopt the Friston-blanket
1084 metaphysical reading. The Goal/Update Coupling Class partition (Class 1 Separated / Class 2 Partial
1085 / Class 3 Coupled), with the explicit Coupled-class scope-exit, is the structural extension that the
1086 Pearl-blanket form invites but does not by itself produce.

1087 **F.7 Relation to existing decompositions**

1088 Zeng and al 2026^[34]’s HalluGuard framework decomposes hallucination risk into *data-driven* and
1089 *reasoning-driven* components, linked respectively to training-time mismatches and inference-time
1090 instabilities. This is the closest-shape decomposition to our $\kappa_{\text{processing}} \times I$ factorization. The structural
1091 axes differ: HalluGuard’s axis is *temporal* (training-time vs inference-time errors); ours is *architectural-*
1092 *vs-informational* (where the goal can flow vs how much it could resolve). The two decompositions live
1093 at different layers and do not reduce to each other. A model can have low data-driven hallucination
1094 (clean training) and still exhibit large goal-coupling displacement (Coupled-class architecture in a
1095 high-ambiguity regime), and vice versa.

1096 Chlon et al. 2025^[33] frame transformers as *Bayesian in expectation but not in realization*, with
1097 permutation-induced order effects scaling as $O(\log n)$. This is structurally distinct from our framing
1098 — we treat the architecture as performing a single goal-conditioned belief update on a fixed input,
1099 not as averaging over orderings — but the two are compatible: their Quantified Martingale Violation
1100 bounds the deviation from Bayesian-in-realization behavior, and our bound captures one specific
1101 channel (goal-coupling) of how non-Bayesian-in-realization a coupled architecture can be.

1102 **F.8 Failed routes — pointer**

1103 Two derivation routes failed at structural levels and are documented in Section A: (F1) Cramér-Rao
1104 inversion fails because Cramér-Rao bounds estimator variance from below, and inverting it for an upper
1105 bound on side-information-induced displacement requires a fixed-estimator-class assumption not
1106 available in coupled-architecture settings. (F2) Rate-distortion inversion fails because rate-distortion
1107 theory describes the minimum bits-per-symbol required to *represent* a source within distortion D ,
1108 not the maximum *displacement induced* by injecting side-information into an update. The pattern:
1109 information-theoretic *lower-bound* machinery for estimator or encoder performance does not invert to
1110 upper bounds on side-information-induced displacement.